

A nondominated sorting simplified swarm optimization with local search mechanisms for multi-objective vehicle routing problems with time windows

Chyh-Ming Lai^a, Chun-Chih Chiu^{b,*}, Tzu-Li Chen^c

^a The Graduate School of Resources Management and Decision Science, National Defense University, Taipei 112, Taiwan

^b Department of Business Management, National Taipei University of Technology, Taipei 106, Taiwan

^c Department of Industrial Engineering and Engineering Management, National Tsing Hua University, Hsinchu 30013, Taiwan

HIGHLIGHTS

- Formulated a new multi-objective model for VRPTW (MO-VRPTW).
- Developed an innovative hybrid multi-objective evolutionary algorithm.
- Integrated LKH for enhanced initial solution generation.
- Proposed an A* -based local search technique.
- Statistical evidence confirmed the algorithm's efficacy for MO-VRPTW.

ARTICLE INFO

Keywords:

Vehicle routing problems with time windows
Multi-objective evolutionary algorithm
Lin–Kernighan heuristic
A* algorithm
MOPSO, NSGA-II
MOEA/D
SPEA2
Nondominated sorting simplified swarm optimization

ABSTRACT

In addressing the complexities of modern logistics, this study introduces a novel multi-objective formulation for vehicle routing problems with time windows (MO-VRPTW), targeting minimizing travel distance, enhancing customer satisfaction, and equalizing driver workloads. We introduce an innovative hybrid multi-objective evolutionary algorithm (MOEA) leveraging nondominated sorting simplified swarm optimization to effectively merge the advantages of various optimization strategies. A key aspect of this advancement is the incorporation of the Lin–Kernighan–Helsgaun (LKH) heuristic, which delivers a superior initial solution, thereby markedly enhancing the speed of convergence. Additionally, we pioneered a local search method inspired by the A* algorithm designed to refine the search process's exploration and exploitation stages. Solomon's benchmark instances, a recognized standard in the VRPTW field, were used to validate our algorithm's effectiveness. Our algorithm demonstrated superior performance in addressing MO-VRPTW through meticulous statistical analysis, outperforming state-of-the-art algorithms, such as MOPSO, NSGA-II, MOEA/D, and SPEA2, regarding efficiency and solution diversity. This study not only advances algorithmic performance but also thoughtfully considers the interests of key supply chain stakeholders.

1. Introduction

In this paper, the focus is on the multi-objective optimization of the vehicle routing problem with time windows (VRPTW), a complex task representative of a multi-objective problem (MOP) that requires simultaneous optimization of two or more conflicting objectives, constrained by specific limitations. Mathematically, a MOP can be defined, without loss of generality, as: $\min f(x) = (f_1(x), f_2(x), \dots, f_p(x))$ subject to

$x \in X \subseteq R^n$, where X is a feasible set in the multi-dimensional space specific to the problem and is defined by $X = \{x \in R^n : g_i(x) \leq 0, i = 1, 2, \dots, m; h_j(x) = 0, j = 1, 2, \dots, n\}$. Here $f(x)$ represents the objective functions, while the constraints are given by the inequality functions $g_i(x)$ and the equality functions $h_j(x)$. Within this context, a solution x dominates another solution y if $\forall i : f_i(x) \leq f_i(y)$ and $\exists j : f_j(x) < f_j(y)$, and it is referred to as Pareto Optimal if no feasible solution dominates it. The goal is to identify the Pareto Set, comprising

* Corresponding author.

E-mail address: ccchiuv@ntut.edu.tw (C.-C. Chiu).

<https://doi.org/10.1016/j.asoc.2025.112989>

Received 14 May 2024; Received in revised form 15 January 2025; Accepted 6 March 2025

Available online 11 March 2025

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all Pareto Optimal solutions, providing decision-makers with a range of compromise options rather than a single solution. This approach is widely applicable in today's complex decision-making landscapes, reflecting the significance of multi-objective optimization [1].

The VRPTW forms the backbone of many logistical and transportation challenges and plays a crucial role in the functionality of supply chain networks. The primary goal of the VRPTW is to determine the optimal set of routes for a fleet of vehicles to service a set of customers within specific time frames, subject to various constraints, such as time windows and vehicle capacities. As a single-objective combinatorial optimization problem, The VRPTW emphasizes the minimization of transportation costs. It represents a complex challenge essential for strategic planning and operational efficiency within supply chain networks, and its NP-hard nature has attracted extensive research efforts [2–4].

Early studies on VRPTW focused on single-objective optimization. However, real-world logistics problems are complex and multidimensional, driving the shift toward multi-objective optimization [5]. Research has expanded from dual-objective models to address a broader range of logistical challenges. Despite these advancements, studies on multi-objective VRPTW (MO-VRPTW) with three or more objectives remain limited. Reference [6] proposed a three-objective model and analyzed trade-offs among these objectives. Ghannadpour and Zarrabi explored three- and four-objective scenarios for homogeneous and heterogeneous fleets, examining trade-offs between cost and customer satisfaction [7]. Ganji et al. introduced a five-objective MO-VRPTW and addressed it using three evolutionary algorithms [8]. Castro-Gutierrez et al. introduced a generalized MO-VRPTW (GMO-VRPTW) with five common objectives. They also proposed a set of problem instances based on real-world data and conducted experiments using NSGA-II on the new dataset and the Solomon dataset. Results demonstrated that the new dataset better captured the characteristics of MO-VRPTW [9]. Zhou and Wang extended the GMO-VRPTW by proposing a local search-based algorithm (LSMOVRPTW). Their results, based on the new dataset and the Solomon dataset, showed superior performance compared to the NSGA-II proposed by Castro-Gutierrez et al. [10]. Srivastava et al. further improved NSGA-II by designing objective-specific crossover and mutation operators. Their experimental results indicated that the improved NSGA-II outperformed LSMOVRPTW in terms of convergence and diversity, particularly for more complex instances [11].

The classical MO-VRPTW primarily focuses on minimizing two often conflicting objectives: the total distance traveled and the number of vehicles used. This dual-objective framework has served as a foundational benchmark for numerous extensions and applications [12–16]. Beyond resource efficiency, workload fairness among drivers has gained increasing attention. Imbalanced route assignments can lead to challenges such as unequal compensation, elevated stress, job dissatisfaction, and adverse health effects caused by long or irregular working hours. In response, researchers have incorporated workload fairness into VRPTW models to balance operational costs with equitable working conditions [17–21]. Customer satisfaction has also emerged as a critical

objective, driven by evolving supply chain demands and heightened customer expectations. For instance, timely or early delivery is particularly essential for perishable goods, making customer-centric objectives a logical progression in VRPTW formulations. These shifts align business operations with consumer needs and enhance competitiveness in dynamic market environments [22–24].

This study introduces a three-objective MO-VRPTW model that addresses three interconnected goals: minimizing total distance traveled, balancing driver workloads, and enhancing customer satisfaction. These objectives align with the priorities of key supply chain stakeholders: the company (cost efficiency), employees (fair working conditions), and customers (timely, reliable deliveries). Specifically, reducing total distance helps control operational costs, balancing workloads promotes fairness and driver well-being, and improving customer satisfaction ensures prompt service for time-sensitive products. By explicitly targeting these objectives, our model reflects the complexity of real-world VRPTW scenarios and provides a stakeholder-focused framework for effective decision-making.

Like the traditional VRPTW, the MO-VRPTW problem is also classified as an NP-hard problem. This complexity implies that exact solution algorithms are only practical for small-scale instances. In multi-objective optimization, several objectives must be considered simultaneously; they cannot be separated or solved sequentially. Consequently, exact methods are rarely suitable for such intricate problems. As a result, researchers have turned to evolutionary algorithms as a solution for MO-VRPTW, leading to a significant number of studies adopting these algorithms [15,18,20,24–26].

This study presents an enhanced version of the nondominated sorting simplified swarm optimization (NSSSO) algorithm, previously recognized for its efficacy in addressing various multi-objective optimization challenges [27]. This version, the improved NSSSO (iNSSSO), incorporates the Lin–Kernighan–Helsgaun (LKH) heuristic [28] for generating high-quality initial solutions and introduces the novel local search technique, A* -based search (ABS), for expansive local search capabilities. The iNSSSO is pivotal for population updates, blending complexity with efficiency to navigate the MO-VRPTW challenges innovatively.

The structure of this study unfolds as follows: Section 2 discusses methods related to solving the MO-VRPTW. Section 3 outlines the specific definitions of the MO-VRPTW adopted in our research. Section 4 delves into the intricacies of the iNSSSO. Experimental findings and their analysis are detailed in Section 5, and Section 6 concludes the study and outlines avenues for future research.

2. Related works

2.1. LKH algorithm

The LKH algorithm is a prominent technique for solving the traveling salesman problem (TSP). Originating in the work of Lin and Kernighan

Table 1
The procedure of SSO.

Initialization phase:	
Step 0:	Initialize x_i^0 randomly, calculate $F(X_i^0)$, define $pBest$ P_i and $gBest$ G , for $i = 1, 2, \dots, nSol$. Let $t = 1$.
Evolution phase:	
Step 1:	Let $i = 1$.
Step 2:	If $tCur > tRun$, halt and output the $gBest$.
Step 3:	Update x_i^t using Eq. (1).
Step 4:	Calculate $F(x_i^t)$ and update P_i (if $F(x_i^t) < F(P_i)$) and G (if $F(P_i) < F(G)$).
Step 5:	If $i < nSol$, let $i = i + 1$ and go back to Step 2 . Otherwise, let $t = t + 1$ and go to Step 1 .

Table 2
The calculation of crowding distance.

Step 1:	For each front ψ_i , $i = 1, 2, \dots, K$ repeat Step 2 and 3 .
Step 2:	Set τ as the count of solutions in ψ_i . For each objective f_j , $j = 1, 2, \dots, M$: Sort ψ_i based on f_j . Set crowding distance of first and last in sorted list to 1, i.e., $cd_j(X_1) = cd_j(X_\tau) = 1$. For the remaining solutions, compute crowding distance as: $cd_j(X_i) = \frac{f_j(X_{i+1}) - f_j(X_{i-1}))}{f_j^{\max} - f_j^{\min}}$
Step 3:	For each solution X_i , compute total crowding distance: $cd(X_i) = \sum_{j=1}^M cd_j(X_i)$.

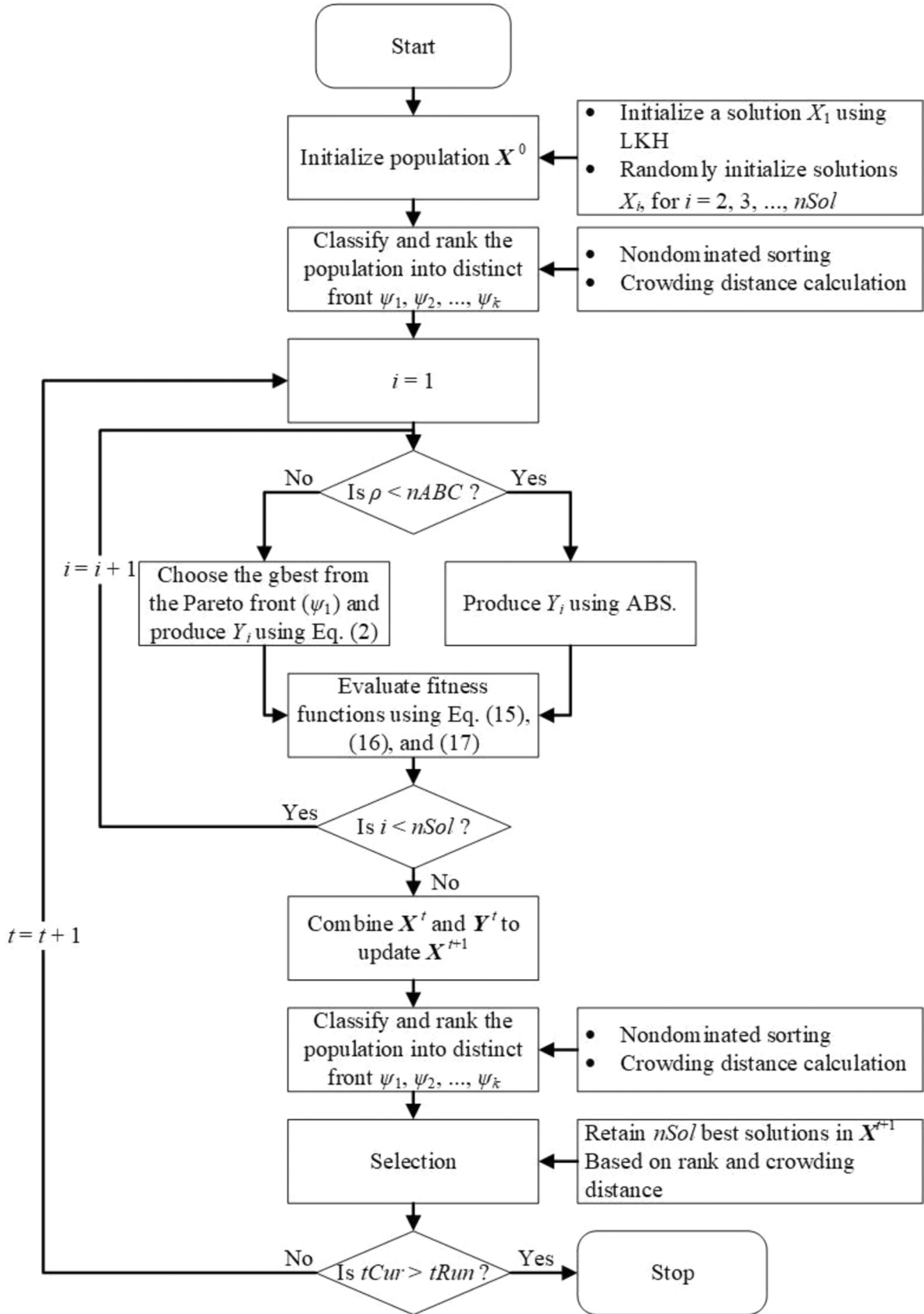


Fig. 1. Flowchart of the proposed iNSSO.

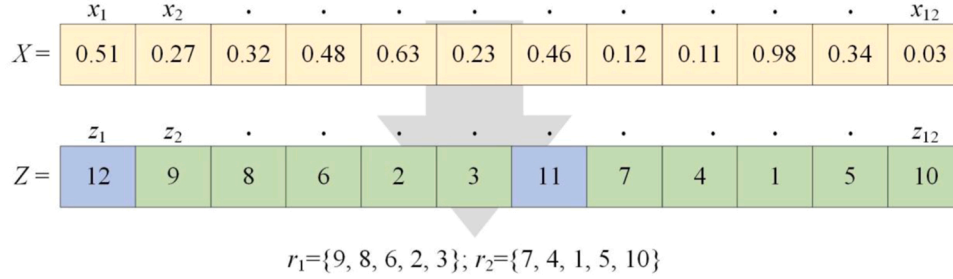


Fig. 2. The solution representation for a MO-VRPTW.

Table 3

The main procedure of ABS.

Step 1:	Initialization
	- Set $open = unassigned = \{c_1, c_2, \dots, c_n\}$, $route = \emptyset$, and $curload = 0$. - Set $curcus = 1$, represents the depot.
Step 2:	Assign customers to the route
	- While $open \neq \emptyset$ - For $i = 1 : open$ $g_{c_i} = g_{curcus} + dt_{curcus, c_i}$ - Compute v_{c_i} and h_{c_i} using $calVcost(c_i)$ and $calHcost(c_i)$, respectively. - Set $validcus = \{c_i v_{c_i} \neq \inf\}$, if $validcus$ is empty, break the loop. - For $i = 1 : validcus$ - Set $gs_{c_i} = \frac{g_{c_i}}{\max_{c_i \in validcus} (g_{c_i})}$, and $hs_{c_i} = \frac{\max_{c_i \in validcus} (h_{c_i}) - h_{c_i}}{\max_{c_i \in validcus} (h_{c_i})}$. - Set $f_{c_i} = gs_{c_i} + hs_{c_i}$. - Apply roulette wheel selection on f to pick next customer c_j. - Update $curcus = c_j$, add $curcus$ to route, and remove $curcus$ from $open$. - Update $curload = curload + dm_{curcus}$. - Calculate $makspan$, considering the return to the depot from $curcus$. - If $makespan \geq avgmsp$, break the loop.
Step 3	Return the route and unassigned customers
	Set $unassigned = open$.

in 1973 [29], it has evolved significantly. LKH iteratively begins with an initial solution and refines it to a local optimum using the k -opt method. During the k -opt process in LKH, the edges added are determined by candidate sets. These sets are constructed using fundamental methods of LKH, like the α -measure and POPMUSIC.

LKH-3 extends the algorithm's capability to tackle more complex TSP variants with constraints, such as limited resources and time windows [28]. By means of transformation techniques [30], LKH-3 can adapt constrained asymmetric TSP and constrained multiple TSP (mTSP) to symmetric forms. LKH-3, with its violation function, explores infeasible solution areas, allowing it to address the VRPTW. To operate the LKH-3 algorithm, two input files are needed:

2.1.1. Problem file

Adhering to the TSPLIB format [31], this file details the VRPTW instance to be addressed. These instances can either be symmetric or

Table 4

The procedure of $calVcost$.

Step 1:	Initialization:
	Set $v = 0$, and $v_1 = v_2 = v_3 = \inf$.
Step 2:	Assess capacity feasibility:
	If adding customer c_i to the route causes the total load to exceed the capacity limit cap, set $v_1 = 0$.
Step 3:	Assess time-window feasibility for customer:
	If the vehicle cannot arrive at customer c_i before $t_{c_i}^l$, set $v_2 = 0$.
Step 4:	Assess time-window feasibility for depot:
	If the vehicle cannot reach the depot before t_{depot}^l after completing service for customer c_i, set $v_3 = 0$.
Step 5:	Calculate overall feasibility:
	If any of v_1, v_2, or v_3 is $\inf$, set $v = \inf$.

Table 5

The procedure of $calHcost$.

Step 1:	Initialization:
	Set $jset = open \setminus \{c_i, prvs = c_i\}$, and $h = 1$.
Step 2:	Main loop
	- While $jset \neq \emptyset$: - Calculate the criteria hf: - For $j = 1 : jset$: - Calculate $d_{c_j} = dt_{prvs, c_j}$, and $t_{c_j} = \frac{t_{c_j}^l}{\max_{c_j \in jset} (t_{c_j}^l)}$. - Calculate $hf_{c_j} = d_{c_j} + t_{c_j}$.
Step 2.1:	Select next customer:
	Set $next = \operatorname{argmin}_{c_j \in jset} (hf_{c_j})$.
Step 2.2:	Assess capacity feasibility:
	If adding customer $next$ to the route causes the total load to exceed the capacity limit cap, exit the loop.
Step 2.3:	Assess time-window feasibility for customer and depot:
	Break the loop if the vehicle cannot meet the time-window constraints for either the customer $next$ or the depot.
Step 2.4:	Update h and loop variables:
	Set $h = h + 1$, $jset = jset$, and $prvs = next$.
Step 2.5:	Return h value

asymmetric. In the asymmetric scenario, LKH converts the instance into a symmetric form with $2n$ vertices.

2.1.2. Parameter file

This includes the problem file's name and certain parameter values that guide the solution process. Any unspecified parameters in this file default to appropriate values.

LKH-3 has shown consistent performance across multiple benchmarks for both the TSP and VRPTW [32–34]. Within this study, LKH-3 is adopted in the proposed iNSSO for the MO-VRPTW to generate initial solutions. This equips iNSSO with feasible solutions from the start and offers potential starting points. For a more comprehensive insight into LKH-3's mechanics and a thorough breakdown of its input parameters, readers can consult the website [35].

2.2. Simplified swarm optimization (SSO)

Simplified swarm optimization (SSO), which serves as the foundation for iNSSO, was introduced by Yeh [25] in 2008. SSO effectively melds swarm intelligence with evolutionary computation. Many studies have highlighted the effectiveness of SSO and its significance across various research domains. The required notations for introducing SSO

Table 6

ANOVA table for parameter setting.

Group	Source	DF	SS	MS	F-value	p-value
IGD _{avg}	C_w	1	0.0248	0.0248	7.87	< 0.0001
	C_g	2	0.7905	0.3953	125.20	< 0.0001
	$nABS$	3	2.4143	0.8048	254.91	< 0.0001

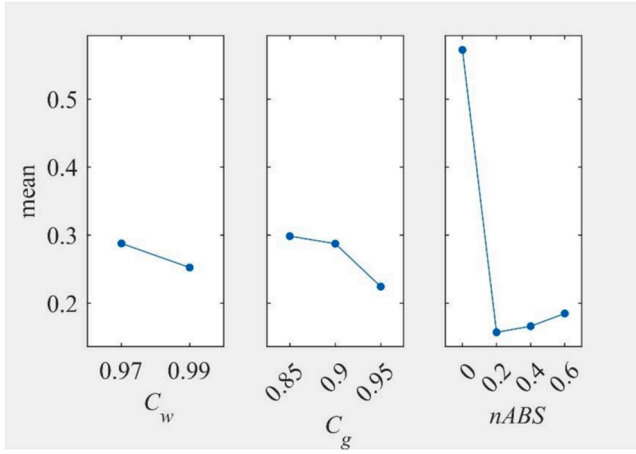


Fig. 3. Main effects plots in average IGD.

are as follows:

$nSol$	The total number of solutions in the population.
$nVar$	The total number of variables in a solution.
$tRun$	The time limit is used as the stop criterion to stop the algorithm.
$tCur$	The current runtime.
X_i^t	$X_i^t = (x_{i,1}^t, x_{i,2}^t, \dots, x_{i,j}^t, \dots, x_{i,nVar}^t)$ represents the i th solution in the population at iteration t , where $i = 1, 2, \dots, nSol$.
P_i	$P_i = (p_{i,1}, p_{i,2}, \dots, p_{i,j}, \dots, p_{i,nVar})$ represents the $pBest$, which is the best solution recorded in the i th solution's evolution history.
G	$G = (g_1, g_2, \dots, g_j, \dots, g_{nVar})$ denotes $gBest$, which is the best solution recorded in the evolution history of all solutions.
$F(X)$	The fitness value of X .
C_g, C_p, C_w	Three predetermined parameters, as defined in [0,1].

As with most prominent evolutionary algorithms, SSO initializes all solutions randomly in its initialization phase. All solutions in the population are encoded as a specific string, which is characterized by a variable vector depending on the problem. During the evolution phase, every solution, denoted as X_i^{t-1} , evolves into a new solution, X_i^t . This is achieved by iteratively adjusting its variable, $x_{i,j}^t$, based on the corresponding variable in $gBest$ (G), its $pBest$ (P_i), its current solution (x_i^{t-1}),

Table 7

Advanced algorithms versus iNSSSO-optimized solutions for instances C1, and C2.

Problem	Best-Known	MODLEM [16]	HSPEA2 [43]	BCO-SIH[45]	N-CLPSO [44]	iNSSSO
C101	828.94	828.94	828.94	828.94	828.94	828.94
C102	828.94	828.94	864.04	828.94	828.94	828.94
C103	828.06	828.94	828.94	835.71	839.35	828.94
C104	824.78	828.94	824.78	885.06	833.67	825.65
C105	828.94	828.94	866.23	916.78	828.94	828.94
C106	828.94	828.94	833.24	828.94	828.94	828.94
C107	828.94	828.94	836.76	828.94	828.94	828.94
C108	828.94	828.94	831.98	831.73	828.94	828.94
C109	828.94	828.94	828.94	840.66	828.94	828.94
C201	591.56	591.56	591.56	591.56	591.56	591.56
C202	591.56	591.56	591.56	591.56	591.56	591.56
C203	591.17	591.17	659.72	593.21	591.17	591.17
C204	590.60	590.60	667.84	606.90	590.60	590.60
C205	588.88	588.88	588.88	588.88	588.88	588.88
C206	588.49	588.49	588.49	588.88	588.49	588.49
C207	588.29	588.29	588.29	590.59	588.26	588.29
C208	588.32	588.32	588.49	593.15	588.32	588.32
Avg	716.13	716.43	729.92	727.67	717.32	716.24
Rank	2.56	2.82	4.12	4.41	2.91	2.76

or a random feasible value (x). The decision among these options is governed by a uniform random number ρ within the interval $[0,1]$ and three predetermined parameters C_g , C_p , and C_w , as delineated in Eq. (1). The stepwise function presented in Eq. (1) is consistently applied throughout each iteration until the stopping criteria are met.

$$x_{i,j}^t = \begin{cases} g_j & \text{if } \rho \in (0, C_g] \\ p_{i,j} & \text{if } \rho \in (C_g, C_p] \\ x_{i,j}^{t-1} & \text{if } \rho \in (C_p, C_w] \\ x & \text{otherwise.} \end{cases} \quad (1)$$

The inherent flexibility of the stepwise function emphasizes the adaptability of SSO, illustrating its potential for easy modifications suitable for different practical applications, as touched upon earlier in this subsection. For instance, in iNSSSO, the $pBest$ from the stepwise function is removed, as shown in Eq. (2). This modification enhances the algorithm's efficiency without compromising solution quality. As a result, based on the no-free-lunch theorem, tailored versions of SSO can be introduced for a range of specific problems. The procedures of the standard SSO for solving a minimization problem are detailed in Table 1.

$$x_{i,j}^t = \begin{cases} g_j & \text{if } \rho \in (0, C_g] \\ x_{i,j}^{t-1} & \text{if } \rho \in (C_g, C_w] \\ x & \text{otherwise.} \end{cases} \quad (2)$$

2.3. NSSSO

The NSSSO is a novel multi-objective evolutionary algorithm (MOEA) introduced by Lai in 2019 [27]. By leveraging the strengths of both swarm intelligence and the nondominated sorting technique based on crowding distance, NSSSO aims to deliver efficient and diversified solutions for MOPs.

2.3.1. Nondominated sorting technique

Nondominated sorting works by categorizing a collection of N solutions $X_1, X_2, \dots, X_N$ into distinct fronts $\psi_1, \psi_2, \dots, \psi_K$. These fronts are organized in decreasing order based on their dominance, adhering to the subsequent criteria:

1. $\forall X_i, X_j \in \psi_k : X_i \not\prec X_j$ and $X_j \not\prec X_i$, $k = 1, 2, \dots, K$. Here $\not\prec$ signifies a nondomination relationship.
2. $\forall X \in \psi_k, \exists X' \in \psi_{k-1} : X' \prec X$, $k = 2, 3, \dots, K$. Here $\prec$ indicates a domination relationship.

Where the front ψ_1 holds the current set of nondominated solutions, displaying the utmost dominance. Following that, ψ_2 represents the second-highest dominance, and so forth. The final front, ψ_K , exhibits the least dominance. Consequently, a solution's nondominance rank in the population can be determined by the sequence of the front to which it belongs.

The nondominated sorting approach implemented in NSSSO utilizes the divide and conquer Algorithm. This algorithm exhibits a time complexity of $O(N \log^{M-1} N)$ in the best case and $O(N^2 \log^{M-2} N)$ in the worst case. Here, M denotes the number of objectives, and N represents the population size. For a comprehensive understanding of the Divide and Conquer Algorithm, readers are encouraged to consult reference [36].

2.3.2. Crowding distance

In the NSSSO algorithm, the crowding distance serves two primary functions. First, it ensures a diverse population, which is essential for covering the Pareto front (PF) comprehensively. Second, it acts as a guiding criterion for selecting the optimal $gBest$, which is crucial for generating a potent new population. To manage selection probabilities

Table 8

Advanced algorithms versus iNSSSO-optimized solutions for instances R1, and R2.

Problem	Best-Known	MODLEM[16]	HSPEA2[43]	BCO-SIH[45]	N-CLPSO[44]	iNSSSO
R101	1645.79	1650.80	1669.81	1643.18	1648.08	1642.88
R102	1486.12	1486.86	1499.00	1476.11	1486.12	1507.35
R103	1292.68	1292.68	1240.08	1245.86	1299.99	1256.16
R104	1007.24	1007.31	1626.47	1026.91	996.27	1041.45
R105	1377.11	1377.11	1391.59	1361.39	1377.11	1367.46
R106	1251.98	1252.03	1261.06	1264.50	1262.80	1280.68
R107	1104.66	1104.66	1127.84	1108.11	1081.17	1095.31
R108	960.88	960.88	990.62	994.68	985.76	982.96
R109	1194.73	1194.73	1197.33	1168.91	1210.73	1215.60
R110	1118.59	1118.84	1125.04	1108.22	1101.49	1164.09
R111	1096.72	1096.73	1090.52	1080.84	1064.67	1113.71
R112	982.14	982.14	982.91	992.22	974.95	1016.29
R201	1252.37	1252.37	1260.58	1197.09	1252.37	1148.83
R202	1191.70	1191.70	1097.58	1092.22	1225.02	1087.37
R203	939.54	939.54	962.40	983.06	962.25	916.20
R204	825.52	731.30	807.29	845.30	766.13	752.04
R205	994.42	965.10	1036.02	999.54	1027.79	969.50
R206	906.14	887.60	941.09	955.94	939.46	910.34
R207	893.33	807.00	892.61	903.59	872.40	831.76
R208	726.75	703.40	805.50	769.96	740.36	718.10
R209	909.16	909.16	938.48	935.57	943.72	878.26
R210	939.34	944.70	981.40	988.34	965.88	933.55
R211	892.71	754.60	854.82	867.95	828.90	776.08
Avg	1086.51	1070.05	1120.87	1087.37	1087.54	1069.82
Rank	3.22	2.76	4.65	3.83	3.50	3.04

and avoid computational issues, crowding distances that traditionally reach infinity, particularly for boundary solutions, are capped at 1. The specific method to calculate crowding distances, initiated after non-dominated sorting, is presented in Table 2.

2.3.3. Framework of NSSSO

NSSSO harmoniously combines swarm intelligence with the strengths of nondominated sorting and crowding distance to deliver efficient multi-objective optimization solutions. The central steps of NSSSO are:

Step	Initialization
1:	- Initialize a population of solutions, X. - Evaluate each solution's fitness for all objectives. - Employ Steps 4 and 5 to classify and rank the population.
Step	Population generation
2:	Formulate a new population, $Y = \{Y_1, Y_2, \dots, Y_{nSol}\}$.
	- For each solution Y_i: - Choose the $gBest$ from the PF (ψ_1). - Construct Y_i using Eq. (2).
Step	Population merging
3:	Combine populations X and Y to update X.
Step	Nondominated sorting
4:	- Apply the sweep line algorithm for nondominated sorting. - Categorize the solutions into distinct fronts $\psi_1, \psi_2, \dots, \psi_K$.
Step	Crowding distance calculation
5:	For each front, calculate the crowding distance of solutions as described in Table 2.
Step	Selection
6:	Retain $nSol$ best solutions in X based on their rank and crowding distance.
Step	Termination Check
7:	If a termination criterion (e.g., max iterations or a time constraint) is met, stop the algorithm. Otherwise, go to Step 2.

3. Mathematical formulation

We use the following notation to give a formal description of the problem:

Parameters	
$G = (N, A)$	A graph with node set N and edge set A .
N	$\{0, 1, 2, \dots, n, n+1\}$ is the node set with 0 and $n+1$ representing the depot and 1 to n representing individual customers.
C	$\{1, 2, \dots, n\}$ is the customer set.
V	$\{1, 2, \dots, m\}$ is the vehicle set.
dm_i	Demand for goods of customer i .
ut_i	Service time due to unloading at customer i .
cp	Capacity of vehicle.
sp_k	Speed of vehicle k .
dt_{ij}	Distance between node i and j .
$[t_i^E, t_i^L]$	Time window of the node i .
tt_{ijk}	Travel time between node i and j by vehicle k calculated as $tt_{ijk} = dt_{ij}/sp_k$.
T_k	The total completion time of the tour for vehicle k .
T_{avg}	The average completion time of the tours across all vehicles used.
Variables	
at_i	Time to arrive at node i , and $at_0 = 0$.
st_i	Time to start servicing node i , $\wedge st_0 = st_{n+1} = 0$.
wt_i	Waiting time of node i due to vehicle arrival later than t_i^E , and $wt_0 = wt_{n+1} = 0$.
vt_i	Time to leave node i , $\wedge vt_0 = vt_{n+1} = 0$.
x_{ijk}	$\begin{cases} 1, & \text{if vehicle } k \text{ travels} \\ & \text{node } i \text{ to } j, \\ 0, & \text{otherwise.} \end{cases}$

In this paper, the VRPTW is characterized by a fleet of homogeneous vehicles, represented by V , a set of customers C , and a directed graph $G = (N, A)$. The graph comprises $n+2$ nodes, where the nodes labeled 1, 2, ..., n correspond to the customers, and the nodes labeled 0 and $n+1$ signify the starting and returning depots, respectively. All vehicles, each with a uniform capacity of cp , are stationed at the depot, ready for the delivery of goods. Each customer i is associated with a specific demand dm_i and must be serviced precisely once by a single vehicle. An additional service time ut_i , which accounts for the loading or unloading of goods, is appended to the route for each customer i . The time window for each customer i is denoted as $[t_i^E, t_i^L]$, characterized by the earliest and latest permissible arrival times. Vehicles are mandated to arrive at the

customers' locations before the latest specified time, t_i^L , but if they arrive later than the earliest permissible time, t_i^E , customer waiting may occur, leading to potential dissatisfaction. Each vehicle is expected to conclude its individual route within the time window allocated to the depot. A feasible solution to the problem constitutes a collection of routes whereby a vehicle embarks from depot 0, traverses through the customer nodes, and ultimately returns to depot $n + 1$, all while adhering to the constraints of capacity and time windows. This paper focuses on the multifaceted decision-making aspects inherent in the MO-VRPTW, aiming to simultaneously fulfill three key objectives: minimizing transportation costs, as presented in Eq. (3); minimizing customer dissatisfaction, as formulated in Eq. (4), and referenced in [24, 37,38]; minimizing the average imbalance among the vehicles, as described in Eq. (5) and referenced in [39]. The problem is mathematically stated as follows:

$$\text{Min}f_1 = \sum_{(i,j) \in A} \sum_{k \in V} dt_{ij} x_{ijk}. \quad (3)$$

$$\text{Min}f_2 = \frac{\sum_{i \in C} wt_i}{n}. \quad (4)$$

$$\text{Min}f_3 = \frac{1}{|\sigma|} \sum_{k \in \sigma} |T_k - T_{\text{avg}}|, \sigma = \left\{ k \mid \sum_{j \in C} x_{ojk} = 1 \right\}. \quad (5)$$

s.t.

$$\sum_{j \in N} \sum_{k \in V} x_{ijk} = 1, \forall i \in C. \quad (6)$$

$$\sum_{j \in N} x_{ojk} = \sum_{i \in N} x_{i,n+1,k} = 1, \forall k \in V. \quad (7)$$

$$\sum_{i \in N} x_{ihk} - \sum_{j \in N} x_{hjk} = 0, \forall h \in C; \forall k \in V. \quad (8)$$

$$\sum_{i \in C} dm_i \sum_{j \in N} x_{ijk} \leq cp, \forall k \in V. \quad (9)$$

$$at_j \geq vt_i + tt_{ijk} - (1 - x_{ijk})M, \forall (i,j) \in A; \forall k \in V. \quad (10)$$

$$t_i^E \leq st_i \leq t_i^L, \forall i \in N. \quad (11)$$

$$st_i = \max\{at_i, t_i^E\}, \forall i \in N. \quad (12)$$

$$wt_i = st_i - t_i^E, \forall i \in N. \quad (13)$$

$$vt_i = st_i + ut_i, \forall i \in N. \quad (14)$$

Eq. (6) stipulates that each customer i must be assigned to exactly

one vehicle. Eqs. (7) and (8) serve as flow constraints, mandating that each vehicle k departs from node 0 once, leaves node i , where $i \in C$, if and only if it enters that node, and subsequently returns to node $n + 1$. Eq. (9) imposes a restriction that each vehicle k cannot be loaded beyond its capacity. The inequality (10) stipulates that each vehicle k cannot arrive at node j before the leave time from customer i , added to the travel time between these two customers if it is travelling from node i to j , where M is a sufficiently large scalar. Eq. (11) ensures that the time to start servicing customer i , denoted as st_i , is compatible with the customer's time window. Eqs. (12), (13), and (14) detail the calculations of st_i , wt_i , and vt_i , respectively.

4. Proposed solution for MO-VRPTW

This paper introduces the iNSSSO, an enhanced algorithm that combines the capabilities of NSSSO, the LKH, and a new local search technique, ABS. The LKH plays a significant role by providing a promising initial solution, giving NSSSO a strong starting point, and enabling faster convergence. Additionally, ABS draws inspiration from the A* algorithm and is used to explore neighboring solutions, enhancing the exploitation capabilities of iNSSSO. More details about iNSSSO will be provided in the forthcoming sections. The flowchart illustrating the proposed iNSSSO can be found in Fig. 1.

4.1. Representation

When applying evolutionary algorithms to optimization problems, the way a solution is represented can significantly impact the algorithm's performance. In this study, we use the random key representation [40], which ensures the feasibility of preserving solutions during the execution of the evolutionary algorithm. A solution encoded with random keys is depicted as a sequence of real numbers, where the rearrangement of positions after sorting corresponds to the permutation order.

Without loss of generality, consider a MO-VRPTW solution where there is a maximum of $|V|$ vehicles available for $|C|$ customers. Let us define $nVar = |V| + |C| - 1$. With this definition, the solution can be

Table 10
Parameter setting for the selected MOEAs.

Method	Parameter
MOPSO	$Pm = [0.7, 0.9, 1]$, $Rm = [0.005, 0.01, 0.05]$
NSGA-II	$Pc = [0.3, 0.5, 0.7]$, $Rm = [0.005, 0.01, 0.05]$
MOEA/D	$T = [2.5, 10]$, $Pc = [0.3, 0.5, 0.7]$, $Rm = [0.05, 0.1, 0.2]$
SPEA2	$Pc = [0.3, 0.5, 0.7]$, $Rm = [0.005, 0.01, 0.05]$

Table 9
Advanced algorithms versus iNSSSO-optimized solutions for instances RC1, and RC2.

Problem	Best-Known	MODLEM[16]	HSPEA2[43]	BCO-SIH[45]	N-CLPSO[44]	iNSSSO
RC101	1696.94	1696.95	1682.02	1637.40	1635.11	1652.72
RC102	1554.75	1554.75	1532.44	1486.85	1503.42	1518.67
RC103	1261.67	1261.67	1356.21	1299.38	1277.99	1353.13
RC104	1135.48	1135.48	1190.46	1200.60	1135.48	1186.17
RC105	1629.44	1629.44	1591.59	1535.80	1542.55	1643.91
RC106	1424.73	1424.73	1441.08	1403.07	1388.70	1448.37
RC107	1230.48	1230.48	1291.70	1230.32	1230.48	1227.86
RC108	1139.82	1139.82	1156.92	1165.17	1157.12	1164.62
RC201	1406.91	1406.94	1345.36	1315.57	1406.91	1269.32
RC202	1367.09	1365.65	1174.23	1169.72	1169.67	1116.03
RC203	1049.62	1049.62	1038.97	1010.74	1082.57	967.78
RC204	798.41	788.30	883.45	890.28	828.61	795.86
RC205	1297.19	1297.65	1237.81	1221.28	1297.19	1174.62
RC206	1146.32	1146.32	1147.15	1097.65	1146.32	1082.43
RC207	1061.14	763.30	1061.85	1024.17	1095.67	1024.15
RC208	828.14	779.60	865.07	864.56	843.28	791.28
Avg	1251.76	1229.42	1249.77	1222.04	1233.82	1213.56
Rank	3.81	3.56	4.44	3.06	3.38	2.75

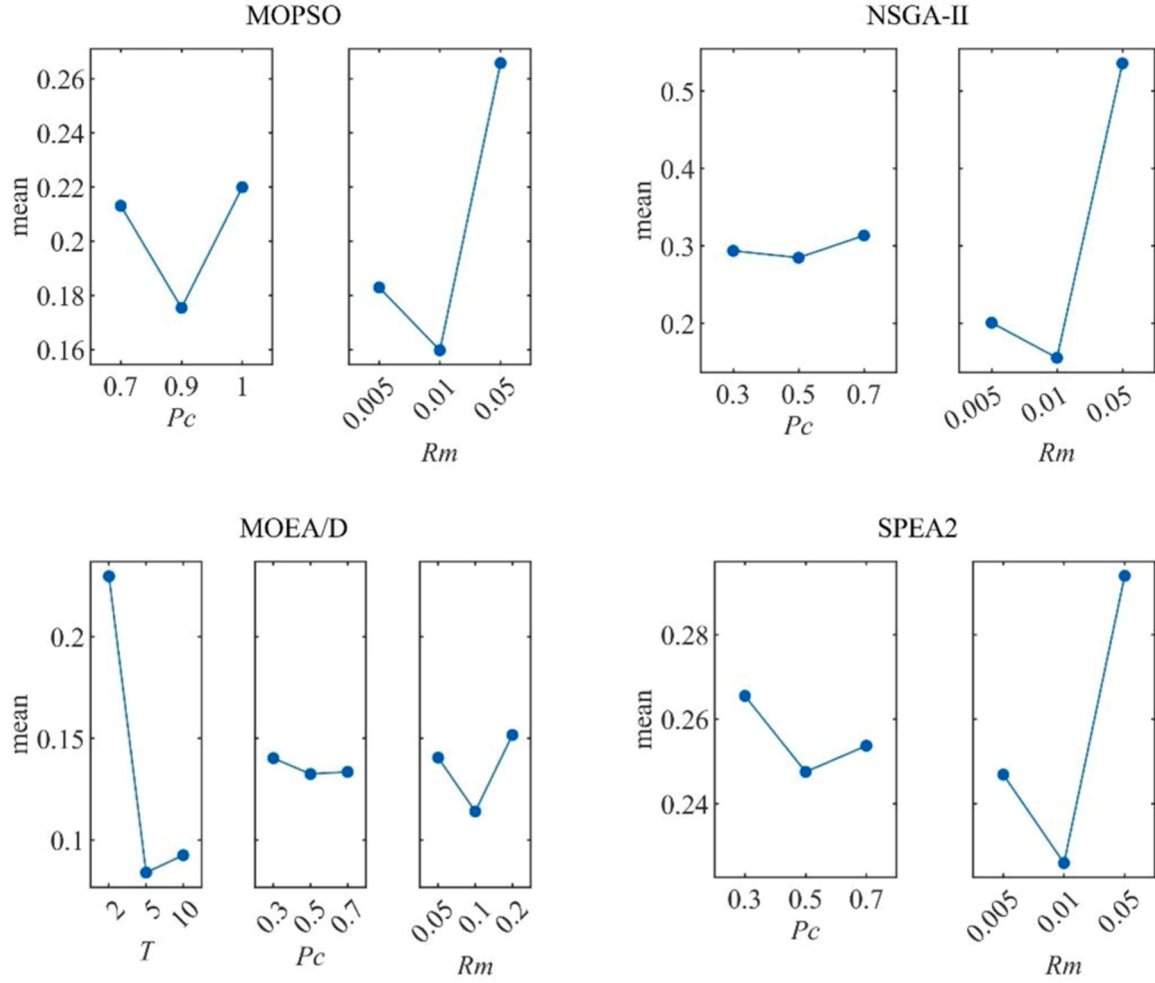


Fig. 4. Main effects plot for the selected MOEAs.

encoded as $X = (x_1, x_2, \dots, x_{nVar}), x_i \in R$. To interpret this encoded solution, each value x_i in X is sorted in ascending order to produce $Z = (z_1, z_2, \dots, z_{nVar}), z_i \in Z$. Values in which $z_i > |C|$ within Z are treated as a route separator. Accordingly, the remaining values where $z_i \leq |C|$ identify individual customers. These are then divided into $|V|$ distinct routes, each starting and finishing at the depot. Within a route, the position of a customer indicates their sequence of visitation by the vehicle. Notably, if the vector Z has consecutive route separators or if separators are positioned at the start or end, it can lead to the creation of empty routes. The collection of non-empty routes represents the specific routing for the MO-VRPTW. A solution representation for a MO-VRPTW with $|V| = 3$ and $|C| = 10$ is illustrated in Fig. 2. This representation comprises two routes.

4.2. Initialization

In the initialization phase of INSSSO, a population of solutions, $X = \{X_1, X_2, \dots, X_{nSol}\}$ is generated, where $nSol$ is the population size. The first solution in this set is derived using the LKH heuristic, while the remaining members of the population are generated randomly. Notably, the LKH heuristic yields a solution expressed as a route set. In order to facilitate iterative updating within the INSSSO framework, it is essential to convert this route-set representation. The conversion process ensures that the LKH-derived solution is transitioned first to an integer-encoded format, Z_1 , and subsequently to its real-number-encoded variant, X_1 . This conversion process involves backtracking from the route-set representation to generate the integer encoding and subsequently deriving

the real-number encoding. Given a route set denoted as $R = \{r_1, r_2, \dots, r_{nRut}\}$, where $nRut$ is the total number of routes within the set, the conversion process can be presented as the following:

Step	Configuration stage
Step 1:	- Define separator set, s, as $\{ C + 1, C + 2, \dots, nVar\}$. - Initialize Z with r_1.
Step 2:	Convert route set to integer code - For i from 2 to $nRut$: - Update Z to $Z \cup s_1 \cup r_i$. - Update separator set s by removing s_1. - Append remaining s to Z.
Step 3:	Convert integer code to real-number code - Initialize X as a $1 \times nVar$ zero matrix. - Generate ρ as a sequence of random real numbers in $[0,1]$ of size $1 \times nVar$, and sorted ρ in ascending order. - For i from 1 to $nVar$: - Set X_{s_i} to ρ_i.

4.3. Solution parsing

Once solutions are generated or updated, it is vital to confirm the feasibility of each solution's route set before evaluating its fitness function. The "Solution Parsing" process serves a dual purpose: primarily, it focuses on ensuring that every customer's service start time aligns with their time window and that the total demand on a route does not breach the vehicle's capacity. Additionally, it calculates each customer's waiting time and each route's total completion time. If any infeasibility

Table 11

Performance results of Solomon's C1 instances.

Data	Criteria	MOPSO	NSGAI	MOEA/D	SPEA2	NSSSO	iNSSSO
C101	Cov	0.07(0.03)	0.14(0.04)	0.09(0.04)	0.14(0.05)	0.07(0.05)	0.59(0.06)
	IGD	144.57(43.66)	77.72(29.12)	78.94(27.58)	108.62(44.79)	1121.38(240.25)	117.74(62.34)
	HV	0.51(0.06)	0.58(0.03)	0.54(0.04)	0.57(0.03)	0.24(0.07)	0.59(0.03)
	Nnds	61.73(15.9)	99.4(0.91)	54.13(8.77)	98.87(2.61)	53.27(12.2)	99.87(0.35)
C102	Cov	0.03(0)	0.09(0.03)	0.1(0.04)	0.02(0.02)	0(0.01)	0.86(0.05)
	IGD	154.97(39.64)	61.79(24.96)	83.22(19.36)	75.4(20.17)	1246.92(83.81)	8.88(6.26)
	HV	0.16(0.04)	0.29(0.05)	0.2(0.04)	0.28(0.04)	0.06(0.05)	0.39(0.04)
	Nnds	42(8.9)	100(0)	56.93(9.83)	100(0)	97.47(4.93)	100(0)
C103	Cov	0.04(0.01)	0.1(0.04)	0.11(0.04)	0.04(0.05)	0(0)	0.82(0.07)
	IGD	270(42.67)	59.54(9.56)	88.45(11.09)	78.74(21.98)	1341.23(150.09)	10.84(7.53)
	HV	0.11(0.03)	0.25(0.04)	0.16(0.03)	0.24(0.04)	0.03(0.04)	0.34(0.04)
	Nnds	21.2(8.99)	100(0)	58(8.87)	100(0)	99.73(0.59)	99.67(1.29)
C104	Cov	0.03(0)	0.14(0.08)	0.09(0.05)	0.09(0.04)	0.01(0.01)	0.74(0.06)
	IGD	299.82(38.46)	63.63(14.73)	122.92(94.45)	74.93(17.71)	1158.84(80.79)	21.1(10.5)
	HV	0.1(0.01)	0.27(0.02)	0.18(0.04)	0.26(0.02)	0.09(0.04)	0.34(0.02)
	Nnds	34(8.5)	99.8(0.77)	84.13(15.99)	100(0)	99.93(0.26)	100(0)
C105	Cov	0.03(0.01)	0.14(0.04)	0.12(0.05)	0.08(0.04)	0.04(0.04)	0.68(0.07)
	IGD	239.92(64.7)	73.06(50.27)	69.47(26.77)	97.76(53.57)	1531.24(133.07)	81.98(71.18)
	HV	0.27(0.13)	0.51(0.08)	0.41(0.1)	0.5(0.07)	0.16(0.11)	0.55(0.07)
	Nnds	49.13(6.57)	100(0)	67.33(9.45)	100(0)	68.33(11.41)	100(0)
C106	Cov	0.03(0)	0.15(0.06)	0.14(0.05)	0.07(0.05)	0.01(0.02)	0.69(0.09)
	IGD	242.57(68.03)	43.23(16.87)	58.82(13.75)	57.43(19.54)	1537.24(128.8)	30.71(21.49)
	HV	0.16(0.05)	0.41(0.06)	0.3(0.05)	0.4(0.07)	0.09(0.09)	0.47(0.06)
	Nnds	29.8(10.6)	99.93(0.26)	68.8(7.54)	100(0)	79.73(15.83)	99.6(1.55)
C107	Cov	0.04(0.03)	0.14(0.04)	0.15(0.07)	0.06(0.04)	0(0.01)	0.7(0.07)
	IGD	149.56(34.99)	38.99(8.37)	56.58(15.8)	58.74(18.76)	1526.69(145.78)	25.79(25.88)
	HV	0.23(0.12)	0.47(0.08)	0.37(0.1)	0.46(0.08)	0.08(0.1)	0.53(0.07)
	Nnds	58.93(12.16)	99.73(1.03)	87.4(15.79)	99.73(1.03)	94.53(8.24)	100(0)
C108	Cov	0.02(0.01)	0.14(0.07)	0.14(0.07)	0.07(0.03)	0(0.01)	0.68(0.08)
	IGD	325.75(69.04)	49.15(16.65)	62.9(11.96)	58.24(21.08)	1530.21(115.32)	23.24(13.69)
	HV	0.15(0.1)	0.43(0.09)	0.31(0.09)	0.42(0.08)	0.09(0.1)	0.49(0.08)
	Nnds	56.87(15.26)	100(0)	80.13(17.88)	100(0)	95.67(5.83)	100(0)
C109	Cov	0.03(0.02)	0.14(0.06)	0.16(0.07)	0.08(0.06)	0(0)	0.67(0.11)
	IGD	195.81(33.15)	42.37(5.83)	69.73(23)	58.23(12.12)	1270.24(147.69)	17.01(7.01)
	HV	0.16(0.1)	0.4(0.06)	0.29(0.08)	0.39(0.05)	0.05(0.04)	0.46(0.05)
	Nnds	62.53(11.19)	99.93(0.26)	87.53(9.01)	100(0)	99.07(2.02)	100(0)
Overall	Cov	0.04(0.01)	0.13(0.02)	0.12(0.03)	0.07(0.03)	0.02(0.02)	0.71(0.08)
	IGD	224.77(67.29)	56.61(13.91)	76.78(20.43)	74.23(18.68)	1362.67(171.77)	37.48(37.12)
	HV	0.21(0.13)	0.4(0.11)	0.31(0.12)	0.39(0.11)	0.1(0.07)	0.46(0.09)
	Nnds	46.24(15.22)	99.87(0.2)	71.6(13.53)	99.84(0.38)	87.53(16.78)	99.9(0.16)

arise during validation, adjustments are made to the solution.

Upon decoding a solution X into a route set, it undergoes the "Solution Parsing" process, where each route in the set is examined individually. For each route, customers are sequentially served. If the service starting time, st_i , exceeds the i th customer's latest permissible arrival time, t_i^l , or if the cumulative demand, including that of the customer, surpasses the vehicle's capacity cap , then customer i is moved to an unassigned customer set, denoted as $unasscus$. Otherwise, the customer remains on the original route. This verification process continues until all customers on the route have been reviewed. Throughout this procedure, key information, such as the waiting time wt_i for each customer i and total completion time T_k for each route, is recorded.

Subsequent to this, if there are still customers in $unasscus$ and the current number of routes in the solution has not reached the maximum $|V|$, new routes will be established for those in $unasscus$. When a new route is added to the route set, customers from $unasscus$ are then sequentially assigned to this new route. Each time a customer is appended, we validate against time window and capacity constraints. If the addition of a customer violates these constraints, another new route is initialized, and we continue adding the remaining customers from $unasscus$ onto this fresh route. This process is reiterated until every customer in $unasscus$ has been allocated to a route or when the total number of routes meets $|V|$. If, after parsing the solution, $unasscus$ still contains unassigned customers, this suggests the solution has maxed out the route limit $|V|$, leaving some customers unserved, rendering the solution infeasible. In such instances, the count of these unserved customers is noted as $restcus$ for solution X . These customers are then

compulsorily added to the final route, the $|V|$ th route. For a feasible solution X , $restcus$ is marked as 0.

4.4. Fitness evaluation

There are three fitness functions, as presented in Eqs. (3)–(5). However, given that certain solutions might be infeasible, we need to incorporate a penalty term within the fitness functions to suppress the fitness values of these infeasible solutions. Specifically, when the number of customers not assigned to a route (denoted as $restcus$) increases, the fitness values of that particular solution should deteriorate, indicating its increasing infeasibility. For a given solution, denoted as X_{sol} , the design of this fitness functions with a penalty term, as shown in Eqs. (15)–(17). The primary purpose behind this design is twofold: firstly, to ensure that feasible solutions receive preference during the optimization phase, and secondly, to assign infeasible solutions a fitness measure that is proportionate to their infeasibility level.

$$f_1(X_{sol}) = \left(\sum_{(i,j) \in A} \sum_{k \in V} dt_{ij} x_{ijk} \right) + restcus_{sol} \times 10^4, \quad (15)$$

$$f_2(X_{sol}) = \frac{\sum_{i \in C} wt_i}{n} + restcus_{sol} \times 10^4, \quad (16)$$

$$f_3(X_{sol}) = \left(\frac{1}{|\sigma|} \sum_{k \in \sigma} (T_{\max} - T_k) / T_{\max} \right) + restcus_{sol} \times 10^4, \quad (17)$$

Table 12
Performance results of Solomon's C2 instances.

Data	Criteria	MOPSO	NSGAI	MOEA/D	SPEA2	NSSSO	iNSSSO
C201	Cov	0.03(0.01)	0.12(0.06)	0.09(0.05)	0.15(0.09)	0.01(0.02)	0.7(0.12)
	IGD	217.83(32.42)	84.63(24.14)	124.26(43.63)	93.86(27.45)	1128.56(142.27)	55.11(29.39)
	HV	0.43(0.09)	0.65(0.05)	0.61(0.05)	0.65(0.05)	0.12(0.1)	0.7(0.04)
	Nnds	85.6(13)	99.8(0.77)	77.33(25.92)	99.6(1.12)	82.27(12.25)	98.67(2.77)
C202	Cov	0.03(0)	0.1(0.03)	0.15(0.07)	0.07(0.09)	0.01(0.01)	0.73(0.1)
	IGD	454.27(82.69)	150.12(20.54)	185.83(48.65)	208.57(34.41)	1372.58(132.61)	48.67(20.14)
	HV	0.36(0.04)	0.52(0.03)	0.43(0.04)	0.52(0.04)	0.22(0.05)	0.64(0.03)
	Nnds	73.87(13.23)	99.93(0.26)	91.73(13.02)	100(0)	99.93(0.26)	99.13(2.85)
C203	Cov	0.03(0.01)	0.1(0.04)	0.04(0.03)	0.13(0.11)	0(0.01)	0.77(0.13)
	IGD	590.09(48.14)	179.95(23.03)	330.07(168.19)	272.25(73.21)	1453.33(114.37)	37.96(25.25)
	HV	0.26(0.03)	0.46(0.03)	0.34(0.06)	0.46(0.03)	0.16(0.04)	0.59(0.02)
	Nnds	72.53(11.67)	99.93(0.26)	98.47(4.88)	100(0)	100(0)	100(0)
C204	Cov	0.02(0)	0.15(0.04)	0.03(0.04)	0.22(0.07)	0.02(0.01)	0.61(0.06)
	IGD	623.8(51)	146.68(20.17)	574.46(512.81)	296.13(54.73)	1271.19(99.83)	53.09(9.38)
	HV	0.27(0.03)	0.46(0.02)	0.31(0.1)	0.43(0.03)	0.21(0.03)	0.54(0.02)
	Nnds	90.73(7.72)	100(0)	98.07(3.58)	99.73(1.03)	100(0)	100(0)
C205	Cov	0.03(0)	0.1(0.04)	0.13(0.07)	0.11(0.05)	0.01(0.01)	0.71(0.11)
	IGD	212.44(35.58)	72.8(10.31)	121.12(51.82)	117.66(32.7)	1187.95(87.39)	50.97(20.47)
	HV	0.44(0.06)	0.65(0.07)	0.57(0.05)	0.65(0.06)	0.2(0.16)	0.71(0.06)
	Nnds	79.67(11.75)	100(0)	90.4(17.75)	100(0)	96.67(4.88)	100(0)
C206	Cov	0.03(0)	0.11(0.03)	0.14(0.07)	0.1(0.06)	0.04(0.04)	0.67(0.09)
	IGD	444.96(77.22)	113.01(37.93)	156.14(107.55)	198.63(55.47)	1271.03(149.88)	66.52(31.53)
	HV	0.26(0.08)	0.65(0.03)	0.56(0.05)	0.64(0.02)	0.33(0.07)	0.72(0.03)
	Nnds	31.87(12.51)	99.93(0.26)	96.8(7.51)	99.87(0.52)	100(0)	100(0)
C207	Cov	0.03(0)	0.12(0.05)	0.15(0.06)	0.14(0.08)	0.03(0.02)	0.61(0.11)
	IGD	405.52(23.58)	106.21(11.03)	118.95(29.92)	181.03(62.06)	1091.62(123.78)	61.11(23.68)
	HV	0.32(0.02)	0.6(0.03)	0.52(0.02)	0.6(0.02)	0.35(0.05)	0.67(0.02)
	Nnds	62.6(6.36)	99.93(0.26)	98.8(2.08)	100(0)	99.8(0.56)	100(0)
C208	Cov	0.03(0)	0.12(0.07)	0.13(0.11)	0.15(0.08)	0(0.01)	0.64(0.13)
	IGD	347.44(69.32)	107.09(14.84)	205.95(135.36)	151.69(39.96)	1264.11(88.29)	51.26(17.88)
	HV	0.35(0.05)	0.61(0.05)	0.5(0.09)	0.63(0.03)	0.2(0.09)	0.69(0.03)
	Nnds	55.93(11.93)	99.93(0.26)	91(14.81)	100(0)	99.53(1.81)	99.87(0.52)
Overall	Cov	0.03(0)	0.12(0.02)	0.11(0.05)	0.13(0.04)	0.01(0.01)	0.68(0.05)
	IGD	388.02(159.2)	111.43(42.52)	209.61(155.71)	175.34(78.98)	1256.73(112.44)	49.08(14.41)
	HV	0.32(0.09)	0.56(0.1)	0.46(0.12)	0.55(0.1)	0.2(0.09)	0.64(0.09)
	Nnds	68.37(17.77)	99.93(0.06)	92.24(6.95)	99.91(0.15)	97.47(5.8)	99.74(0.49)

$$where \sigma = \left\{ k \mid \sum_{j \in C} x_{ojk} = 1 \right\}.$$

4.5. ABS

ABS is a specialized local search mechanism designed specifically for the MO-VRPTW. It inherits the balanced approach of the A* algorithm in pathfinding, optimizing both speed and accuracy. When constructing routes, ABS takes into account not only the feasibility of the routes but also aims to balance the makespan across different routes. By applying a probabilistic route selection strategy and using the roulette wheel selection method, ABS is capable of discovering diverse and promising routes, thereby enhancing the efficacy of iNSSSO in a local search.

The decision to engage ABS for each solution is controlled by a predefined parameter $nABS$ that ranges between 0 and 1, along with a randomly generated number. If the random number is less than $nABS$, the solution proceeds to the local search phase. Once ABS is activated, solution X initiates a local search by randomly selecting $\max\left(1, \text{round}\left(\frac{|C|}{3}\right)\right)$ routes from its existing set of routes, denoted as τ . Preference is given to routes serving a smaller number of customers, $nCus_k$, as they are more likely to be efficiently consolidated into a reduced number of more efficient routes. The selection probability, P_k , for each route k in the set τ , based on the number of customers $nCus_k$ it serves, is given by Eq. (18), and the roulette wheel selection method is used for the route selection:

$$P_k = \frac{P_k}{\sum_{k \in \tau} P_k}, \text{ where } p_k = e^{\frac{-10nCus_k}{\max_{k \in \tau}(nCus_k)}} \quad (18)$$

BSet τ , and their customers are transferred to the set denoted as *unassigned*. ABS is then activated to redistribute these unassigned customers into new routes. The main procedure of ABS for individual route construction is outlined in Table 3. If any customers remain unassigned in *unassigned* after ABS has run its course, ABS is triggered again to create another new route. This process continues until *unassigned* is empty or the number of routes hits the maximum allowable vehicle count. As detailed in Step 2 of Table 3, ABS applies three stopping criteria while constructing each individual route: 1) when the set *open* is empty, 2) when the set *validcus* is empty, or 3) when the makespan of the currently building route is greater than or equal to the average makespan (*avgmsp*) of all routes in the current solution.

Starting with a set of unassigned customers, ABS builds a vehicle route by using two key metrics: v_{c_i} and h_{c_i} calculated by *calVcost* and *calHcost*, as shown in Tables 4 and 5, respectively. The metric v_{c_i} assesses the feasibility of adding a specific customer c_i to a route, considering factors like vehicle capacity and time-windows for both the customer and the depot. The metric h_{c_i} evaluates the customer c_i 's potential for further route extension by estimating how many additional customers could be added without violating constraints. Customers are scored and selected using a composite feasibility criterion f , and the route-building process continues until one of the three aforementioned stopping criteria is met.

5. Experimentation and discussion

In this section, we validate the effectiveness and rationality of iNSSSO through a comprehensive experimental analysis. Section 5.1 introduces the metrics applied to assess the performance of iNSSSO. In Section 5.2, we detail the parameter settings for our algorithm, focusing on the process used to optimize them. Section 5.3 presents a

Table 13
Performance results of Solomon's R1 instances.

Data	Criteria	MOPSO	NSGAI	MOEA/D	SPEA2	NSSSO	iNSSSO
R101	Cov	0(0)	0.03(0.03)	0(0)	0.05(0.05)	0.01(0.02)	0.92(0.05)
	IGD	62.24(32.29)	12.35(9.53)	23.07(11.03)	12.63(10.46)	1460.56(3612.5)	2.9(7.84)
	HV	0.05(0.03)	0.08(0.03)	0.05(0.02)	0.09(0.04)	0.01(0.02)	0.12(0.04)
	Nnds	16.87(7.55)	49(8.97)	18(6.47)	93.93(7.09)	17.87(5.96)	99.87(0.35)
R102	Cov	0.04(0.01)	0.05(0.01)	0.05(0.01)	0.03(0.03)	0.01(0.02)	0.96(0.03)
	IGD	78.93(31.19)	20.35(12.71)	31.09(13.66)	17.64(10.51)	672.43(114.42)	3.05(5.94)
	HV	0.04(0.02)	0.1(0.03)	0.05(0.02)	0.1(0.03)	0.01(0.02)	0.16(0.04)
	Nnds	7.4(2.16)	98.4(2.53)	27.93(7.43)	99.73(0.8)	58.27(9.75)	100(0)
R103	Cov	0.04(0)	0.08(0.04)	0.06(0.02)	0.01(0.02)	0(0.01)	0.92(0.04)
	IGD	111.73(88.88)	16.9(11.29)	31.14(20.14)	22.16(8.86)	978(112.73)	3.01(7.39)
	HV	0.04(0.02)	0.11(0.03)	0.06(0.02)	0.11(0.03)	0.01(0.03)	0.18(0.04)
	Nnds	50.87(14.07)	99.6(1.3)	46.4(9.31)	100(0)	81.2(14.57)	99.87(0.52)
R104	Cov	0.03(0.01)	0.09(0.04)	0.05(0.02)	0.02(0.03)	0.01(0.03)	0.9(0.07)
	IGD	195.43(85.32)	22.53(17.24)	37.32(20.83)	32.36(21.06)	1011.55(74.82)	7.98(23.97)
	HV	0.07(0.02)	0.16(0.03)	0.1(0.02)	0.15(0.02)	0.01(0.02)	0.22(0.03)
	Nnds	28.53(12.32)	100(0)	74.53(11.45)	100(0)	94.13(7.48)	99.93(0.26)
R105	Cov	0.02(0.02)	0.11(0.06)	0.03(0.02)	0.11(0.05)	0.01(0.02)	0.79(0.1)
	IGD	72.74(50.07)	15.58(10.81)	23.35(9.99)	15.88(11.12)	632.72(60.33)	6.67(8.27)
	HV	0.05(0.02)	0.16(0.03)	0.09(0.02)	0.16(0.03)	0.04(0.03)	0.2(0.03)
	Nnds	53.6(13.43)	98.33(2.64)	44.53(8.92)	99.67(0.72)	33.8(7.08)	100(0)
R106	Cov	0.04(0)	0.05(0.01)	0.04(0.01)	0.03(0.03)	0(0.01)	0.96(0.03)
	IGD	93.61(61.5)	13.96(3.54)	25.79(5.42)	18.48(5.47)	747.86(92.76)	1.58(3)
	HV	0.04(0.02)	0.13(0.03)	0.06(0.02)	0.14(0.03)	0.02(0.02)	0.2(0.03)
	Nnds	49.87(10.27)	99.93(0.26)	37(7.87)	99(2.3)	82.27(10.55)	100(0)
R107	Cov	0.04(0)	0.07(0.03)	0.06(0.03)	0.02(0.02)	0(0)	0.92(0.04)
	IGD	168.48(96.43)	16.46(1.4)	24.41(2.97)	26.74(7.93)	940.23(67.33)	0.98(0.7)
	HV	0.06(0.02)	0.16(0.02)	0.09(0.01)	0.15(0.02)	0.01(0.01)	0.23(0.02)
	Nnds	60.27(11.03)	100(0)	74.87(13.54)	100(0)	90.67(10.77)	100(0)
R108	Cov	0.03(0.01)	0.14(0.04)	0.05(0.02)	0.07(0.04)	0(0)	0.79(0.06)
	IGD	168.36(83.63)	15.77(3.63)	37.74(28.49)	24.09(6.74)	1024.27(100.87)	4.67(4.82)
	HV	0.09(0.02)	0.19(0.03)	0.12(0.02)	0.18(0.03)	0.02(0.03)	0.24(0.03)
	Nnds	21.53(12.93)	100(0)	62.2(13.07)	100(0)	94.8(8.93)	99.93(0.26)
R109	Cov	0.03(0)	0.19(0.09)	0.04(0.01)	0.18(0.1)	0(0.01)	0.65(0.14)
	IGD	74.38(64.26)	9.92(3.77)	19.46(4.99)	12.91(6.39)	758.61(88.81)	5.37(5.2)
	HV	0.06(0.02)	0.17(0.02)	0.08(0.01)	0.17(0.02)	0.02(0.03)	0.19(0.03)
	Nnds	25.73(7.4)	99.27(1.62)	50.07(9)	100(0)	46.93(9.82)	99.93(0.26)
R110	Cov	0.03(0.01)	0.16(0.08)	0.06(0.02)	0.14(0.05)	0(0)	0.71(0.1)
	IGD	183.43(131.16)	8.85(2.11)	27.92(4.36)	14.07(5.99)	919.05(128.01)	3.67(2.17)
	HV	0.06(0.02)	0.16(0.02)	0.08(0.01)	0.16(0.02)	0.01(0.01)	0.2(0.02)
	Nnds	42.93(12.16)	100(0)	29.33(3.6)	100(0)	55.33(17.93)	100(0)
R111	Cov	0.06(0.03)	0.09(0.03)	0.06(0.02)	0.05(0.03)	0.01(0.02)	0.83(0.05)
	IGD	146.4(116.9)	42.33(33.31)	50.82(30.72)	52.64(39.2)	911.51(106.59)	32.64(27.31)
	HV	0.26(0.07)	0.38(0.08)	0.3(0.08)	0.37(0.08)	0.14(0.04)	0.44(0.06)
	Nnds	57.8(12.68)	100(0)	52.93(8)	100(0)	76.87(17.49)	99.87(0.35)
R112	Cov	0.19(0.04)	0.14(0.05)	0.07(0.02)	0.11(0.07)	0.01(0.01)	0.57(0.08)
	IGD	184.14(70.68)	72.05(23.91)	72.7(22.56)	69.5(24.49)	746.26(76.23)	114.93(35.89)
	HV	0.31(0.02)	0.41(0.02)	0.35(0.02)	0.4(0.02)	0.17(0.03)	0.46(0.02)
	Nnds	80(12.04)	99.93(0.26)	55.6(9.18)	100(0)	81.87(10.34)	99.93(0.26)
Overall	Cov	0.05(0.05)	0.1(0.05)	0.05(0.02)	0.07(0.05)	0(0)	0.83(0.13)
	IGD	128.32(50.89)	22.25(17.92)	33.73(14.96)	26.59(17.52)	900.25(221.63)	15.62(32.4)
	HV	0.09(0.09)	0.18(0.1)	0.12(0.1)	0.18(0.1)	0.04(0.05)	0.24(0.1)
	Nnds	41.28(21.3)	95.37(14.62)	47.78(17.84)	99.36(1.73)	67.83(25.13)	99.94(0.06)

comparative analysis of our results against previously published studies and other prevalent MOEAs, accompanied by a brief discussion. To further substantiate the superiority of iNSSSO in solving the MO-VRPTW, a nonparametric statistical test is performed and discussed in [Section 5.4](#). Initially, we provide an overview of the benchmark instances used for comparison.

Because there are no established benchmarks specifically designed for the MO-VRPTW, we tested the proposed iNSSSO algorithm using standard Solomon's benchmark datasets, as referenced in [\[41,42\]](#). These datasets offer problem sets of varying sizes, classified by the number of customers: 25, 50, and 100. Each size category contains 56 problem sets further classified into six groups: C1, R1, RC1, C2, R2, and RC2, based on criteria such as the width of the time window and the spatial pattern's aggregation degree. This study zeroes in on the 100-customer problem sets, each enforcing a cap of 25 vehicles. The data for each set sequentially lists customer details, including index, coordinates (x, y), demand, service window (ready time and due date), and service duration. Travel times between customers are computed using their Euclidean distance.

All experiments were conducted in MATLAB, running on an AMD Ryzen 3900X GHz CPU with 64 GB of RAM, using the Microsoft Windows 10 operating system. The computational runtime is measured in CPU seconds.

5.1. Performance metrics

Evaluating the performance of MOEAs is inherently more challenging than assessing single-objective algorithms. Although various metrics have been proposed, no single criterion has gained universal acceptance for evaluating the overall effectiveness of MOEAs. In this study, we apply four widely recognized metrics to assess the performance of our proposed iNSSSO algorithm. Let P be a set of uniformly distributed solutions on the true PF, and let P be the set of nondominated solutions discovered by the MOEA. The four metrics we use are described as follows:

Table 14

Performance results of the Solomon's R2 instances.

Data	Criteria	MOPSO	NSGAII	MOEA/D	SPEA2	NSSSO	iNSSSO
R201	Cov	0.02(0.01)	0.17(0.09)	0.04(0.02)	0.18(0.08)	0.07(0.05)	0.59(0.08)
	IGD	315.61(97.56)	64.13(41.46)	65.79(13.29)	84.28(44.91)	620.41(70.54)	41.31(21.18)
	HV	0.1(0.04)	0.36(0.04)	0.21(0.03)	0.37(0.04)	0.21(0.04)	0.42(0.04)
	Nnds	8.8(6.74)	99.67(0.82)	52.93(12.91)	99.93(0.26)	93.93(7.64)	99.53(1.81)
R202	Cov	0.04(0.01)	0.06(0.03)	0.05(0.02)	0.01(0.02)	0.01(0.01)	0.94(0.04)
	IGD	311.16(62.5)	69.04(10.2)	105.54(39.1)	90.77(18.96)	802.19(79.13)	8.7(7.08)
	HV	0.08(0.03)	0.28(0.04)	0.16(0.04)	0.3(0.04)	0.12(0.03)	0.45(0.03)
	Nnds	27.27(8)	100(0)	65.8(12.37)	99.87(0.52)	99.53(1.36)	99.73(1.03)
R203	Cov	0.03(0)	0.16(0.08)	0.11(0.03)	0.1(0.05)	0.01(0.01)	0.68(0.09)
	IGD	379.72(56.84)	56.24(6.64)	85.39(10.24)	91.46(29.44)	929.39(114.65)	19.78(6.86)
	HV	0.13(0.03)	0.34(0.02)	0.24(0.03)	0.33(0.03)	0.11(0.03)	0.45(0.03)
	Nnds	53.2(9.92)	100(0)	85.53(11.84)	100(0)	100(0)	100(0)
R204	Cov	0.03(0)	0.1(0.05)	0.08(0.03)	0.15(0.06)	0.05(0.04)	0.68(0.07)
	IGD	403.52(54.93)	71.94(8.66)	96.38(17.57)	166.29(56.55)	746.46(68.06)	38.17(23.53)
	HV	0.15(0.03)	0.36(0.03)	0.3(0.05)	0.36(0.03)	0.18(0.04)	0.48(0.03)
	Nnds	74.47(13.54)	100(0)	94.13(6.81)	100(0)	100(0)	100(0)
R205	Cov	0.02(0)	0.17(0.07)	0.06(0.03)	0.28(0.08)	0.01(0.01)	0.53(0.11)
	IGD	317.42(78.12)	41.95(8.11)	54.84(7.94)	61.22(16.24)	769.92(61.48)	22.97(12.19)
	HV	0.11(0.03)	0.37(0.04)	0.25(0.04)	0.39(0.04)	0.15(0.05)	0.43(0.03)
	Nnds	33(8.06)	99.93(0.26)	89.4(12.26)	99.8(0.56)	100(0)	100(0)
R206	Cov	0.03(0.01)	0.11(0.04)	0.07(0.04)	0.02(0.02)	0(0.01)	0.86(0.05)
	IGD	373.7(55.17)	62.26(14.76)	93.6(24.78)	100.11(25.28)	857.33(64.17)	8.57(6.81)
	HV	0.11(0.03)	0.36(0.03)	0.24(0.03)	0.34(0.03)	0.13(0.04)	0.48(0.02)
	Nnds	10.8(2.48)	100(0)	64.07(14.57)	100(0)	100(0)	100(0)
R207	Cov	0.03(0)	0.11(0.07)	0.05(0.03)	0.17(0.09)	0.01(0.01)	0.71(0.11)
	IGD	426.66(33.35)	63.47(8.53)	105.67(17.26)	105.24(21.3)	883.4(103.04)	22.62(9.7)
	HV	0.1(0.02)	0.38(0.04)	0.24(0.03)	0.39(0.04)	0.16(0.04)	0.5(0.03)
	Nnds	78.6(15.8)	100(0)	97.53(4)	100(0)	99.93(0.26)	100(0)
R208	Cov	0.02(0.01)	0.16(0.04)	0.07(0.03)	0.17(0.06)	0.1(0.05)	0.54(0.06)
	IGD	378.34(29.19)	61.06(9.54)	104.97(38.13)	209.64(47.59)	714.13(57.71)	57.58(34.83)
	HV	0.13(0.02)	0.39(0.03)	0.28(0.03)	0.38(0.03)	0.23(0.04)	0.48(0.02)
	Nnds	85.13(13.96)	100(0)	96.6(4.4)	100(0)	100(0)	100(0)
R209	Cov	0.03(0)	0.17(0.08)	0.05(0.02)	0.15(0.09)	0.02(0.02)	0.67(0.08)
	IGD	473.1(104.75)	48.28(28.27)	82(81.17)	119.11(58.52)	803.58(94.32)	16.16(4.4)
	HV	0.13(0.02)	0.39(0.03)	0.28(0.03)	0.37(0.02)	0.2(0.03)	0.46(0.02)
	Nnds	16.2(3.99)	99.93(0.26)	89.93(12.29)	100(0)	100(0)	99.93(0.26)
R210	Cov	0.03(0)	0.13(0.05)	0.07(0.04)	0.09(0.06)	0.01(0.01)	0.76(0.08)
	IGD	495.44(98.75)	50.62(11.5)	72.29(18.39)	99.91(37.43)	803(88.76)	12.64(4.42)
	HV	0.1(0.02)	0.33(0.03)	0.22(0.03)	0.33(0.03)	0.13(0.04)	0.42(0.03)
	Nnds	30.47(11.13)	99.8(0.77)	89.2(9.09)	100(0)	100(0)	99.8(0.56)
R211	Cov	0.02(0)	0.19(0.08)	0.06(0.03)	0.24(0.09)	0.03(0.02)	0.53(0.1)
	IGD	459.2(42.58)	51.78(17.6)	99.69(96.34)	100.76(31.05)	896.54(98.54)	23.55(6.84)
	HV	0.12(0.03)	0.44(0.03)	0.3(0.04)	0.45(0.03)	0.22(0.04)	0.5(0.02)
	Nnds	41.93(10.46)	100(0)	91.87(10.98)	99.93(0.26)	99.67(1.29)	99.87(0.52)
Overall	Cov	0.04(0.05)	0.14(0.04)	0.06(0.02)	0.14(0.08)	0.03(0.03)	0.67(0.13)
	IGD	376.5(86.37)	59.4(9.69)	86.57(17.16)	108.19(41.39)	797.72(86.63)	32.25(29.79)
	HV	0.13(0.06)	0.37(0.04)	0.26(0.05)	0.37(0.04)	0.17(0.04)	0.46(0.03)
	Nnds	44.99(28.43)	99.94(0.1)	81.05(16.5)	99.96(0.07)	97.91(5.34)	99.9(0.15)

1. Coverage (Cov): This metric is designed to assess how effectively the set of nondominated solutions P discovered by a MOEA covers P . Specifically, the coverage metric $Cov(P, P)$ is defined as the ratio of the number of solutions in P that are covered by P , to the total number of solutions in P .

$$Cov = \frac{|\{v \in P; \exists v \in P : v \preceq v\}|}{|P|} \quad (19)$$

In this equation, $v \preceq v$ indicates that solution v covers v , which occurs if v either dominates v or is identical to v in terms of objectives. A Cov value of 1 signifies that all points in P are either dominated by or equal to points in P . Conversely, a Cov value of 0 indicates that none of the points in P are covered by P .

2. Inverted generation distance (IGD): IGD is commonly used to assess both convergence and diversity. It is defined by the equation:

$$IGD = \frac{\sum_{v \in P} d(v, P)}{|P|}, \quad (20)$$

where $d(v, P)$ denotes the minimum Euclidean distance between v and the points in P . A low IGD value indicates that the set P closely

approximates P and covers its entire range. Therefore, a smaller IGD value is generally considered to represent better algorithmic performance.

3. Hypervolume (HV): HV measures the volume between all solutions in a nondominated set P and a nadir point, which represents the worst objective values obtained by P . To calculate the HV, a set of hypercubes c_i is formed, with each solution and the nadir point serving as diagonal corners. HV is then the total volume of all these hypercubes and calculated as Eq. (21). Higher HV values indicate better convergence and diversity of the solutions on the Pareto front.

$$HV = volume(i = 1 || P | c_i) \quad (21)$$

In this study, the calculated HV values are too large for effective result presentation. To address this, the objective values f_i of each nondominated solution are scaled relative to the nadir point, defined as $(\max(f_1(P)), \max(f_2(P)), \dots, \max(f_M(P)))$, where $\max(f_i(P))$ is the worst objective value for each objective in P . The transformed objective value f'_i is calculated as $f'_i = f_i / \max(f_i(P))$, normalizing the values. HV is then computed using these transformed objective values, making the results more manageable and easier to present.

Table 15

Performance results of the Solomon's RC1 instances.

Data	Criteria	MOPSO	NSGAI	MOEA/D	SPEA2	NSSSO	iNSSSO
RC101	<i>Cov</i>	0.03(0.02)	0.07(0.04)	0.03(0.02)	0.11(0.07)	0(0)	0.84(0.09)
	<i>IGD</i>	55.51(14.58)	10.79(4.91)	22.65(8.64)	8.89(2.27)	803.46(108.86)	1.4(0.9)
	<i>HV</i>	0.06(0.05)	0.13(0.04)	0.08(0.04)	0.13(0.05)	0.01(0.01)	0.17(0.04)
	<i>Nnds</i>	71.27(11.94)	89.2(12.92)	33.87(9.5)	98.8(2.24)	29.33(6.84)	99.6(1.12)
RC102	<i>Cov</i>	0.04(0.01)	0.05(0.03)	0.04(0.01)	0.02(0.03)	0(0)	0.97(0.02)
	<i>IGD</i>	87.77(33.93)	12.72(1.16)	29.91(10)	20.08(6.41)	1016.37(127.01)	0.57(1.32)
	<i>HV</i>	0.03(0.02)	0.1(0.03)	0.04(0.02)	0.11(0.03)	0(0.01)	0.18(0.03)
	<i>Nnds</i>	54.87(11.21)	99.33(1.91)	35.07(7.46)	99.8(0.56)	43.73(10.81)	99.93(0.26)
RC103	<i>Cov</i>	0.04(0)	0.05(0.01)	0.07(0.02)	0.02(0.02)	0(0)	0.96(0.03)
	<i>IGD</i>	160.44(124.09)	15.96(1.41)	40.61(49.61)	22.6(5.15)	1231.18(137.4)	0.37(0.24)
	<i>HV</i>	0.04(0.01)	0.1(0.02)	0.05(0.01)	0.1(0.02)	0(0)	0.18(0.02)
	<i>Nnds</i>	45.67(9.4)	100(0)	47.2(13.16)	99.93(0.26)	60.4(15.4)	100(0)
RC104	<i>Cov</i>	0.03(0.01)	0.12(0.04)	0.1(0.04)	0.04(0.03)	0(0)	0.79(0.05)
	<i>IGD</i>	223.94(77.63)	16.59(2.73)	35.82(13.23)	27.05(8.41)	1407.49(146.28)	3.52(2.14)
	<i>HV</i>	0.08(0.02)	0.17(0.02)	0.11(0.02)	0.16(0.02)	0.01(0.02)	0.22(0.02)
	<i>Nnds</i>	31(10.93)	99.93(0.26)	52.07(8.81)	100(0)	77.2(12.25)	100(0)
RC105	<i>Cov</i>	0.02(0.01)	0.03(0.04)	0.02(0.02)	0.03(0.03)	0(0.01)	0.94(0.04)
	<i>IGD</i>	133.73(83.39)	13.06(5.56)	34.56(14.31)	14.4(5.23)	1025.45(113.94)	2.26(5.56)
	<i>HV</i>	0.02(0.01)	0.1(0.02)	0.04(0.02)	0.1(0.03)	0(0.01)	0.16(0.03)
	<i>Nnds</i>	27.6(7.51)	97.2(5.63)	25.53(6.64)	98.93(2.84)	31.33(11.3)	100(0)
RC106	<i>Cov</i>	0.05(0.02)	0.09(0.04)	0.04(0.01)	0.06(0.04)	0(0)	0.87(0.06)
	<i>IGD</i>	86.8(67.31)	11.18(2.19)	25.13(9.27)	20.14(6.02)	869.75(119.29)	5.28(8.09)
	<i>HV</i>	0.13(0.09)	0.22(0.09)	0.16(0.09)	0.22(0.08)	0.02(0.03)	0.26(0.08)
	<i>Nnds</i>	71.53(11.86)	99.73(1.03)	41.4(8.64)	99.8(0.77)	44.07(14.56)	99.73(1.03)
RC107	<i>Cov</i>	0.04(0.03)	0.1(0.06)	0.04(0.03)	0.1(0.06)	0(0)	0.8(0.08)
	<i>IGD</i>	276.07(169.41)	18.79(9.23)	42.6(37.12)	25.75(11.79)	1194(137.31)	12.27(13.82)
	<i>HV</i>	0.15(0.11)	0.27(0.11)	0.2(0.12)	0.27(0.11)	0.04(0.05)	0.32(0.11)
	<i>Nnds</i>	71.87(9.24)	99.87(0.52)	51.53(12.73)	99.73(1.03)	47.33(13.43)	99.73(1.03)
RC108	<i>Cov</i>	0.09(0.05)	0.11(0.06)	0.06(0.02)	0.08(0.04)	0(0)	0.75(0.08)
	<i>IGD</i>	238.31(135.33)	55.19(38.98)	62.41(34.82)	57.87(39.71)	1121.86(139.56)	53.08(40.64)
	<i>HV</i>	0.25(0.07)	0.38(0.07)	0.32(0.07)	0.38(0.06)	0.12(0.05)	0.44(0.06)
	<i>Nnds</i>	72.4(10.72)	100(0)	53.2(13.03)	100(0)	63.67(12.61)	100(0)
Overall	<i>Cov</i>	0.05(0.05)	0.11(0.05)	0.05(0.02)	0.09(0.06)	0.01(0.01)	0.79(0.14)
	<i>IGD</i>	239.54(156.85)	31.42(22.2)	51.7(25.19)	48.84(39.17)	993.24(208.98)	20.5(33.24)
	<i>HV</i>	0.12(0.09)	0.26(0.13)	0.18(0.11)	0.25(0.13)	0.08(0.08)	0.31(0.14)
	<i>Nnds</i>	51.23(21.89)	98.74(3.11)	55.54(22.82)	99.74(0.42)	64.88(26.49)	99.88(0.13)

4. Number of nondominated solutions (*Nnds*): This metric counts the nondominated solutions found by a MOEA. A higher *Nnds* value suggests better search capability and greater solution diversity. However, a high count may require more computational resources. This metric is most informative when considered alongside other performance indicators like *Cov*, *IGD*, and *HV*.

The calculation of the metrics previously mentioned, like *IGD*, *Cov* and *HV*, rely on knowledge of the true PF. However, as the true PF is unknown for the test cases examined in this study, it is estimated for each problem scenario.

5.2. Parameter settings

In the proposed iNSSSO algorithm, several parameters, including C_w , C_g , and $nABS$, significantly influence the algorithm's performance. Consequently, optimal values for these parameters must be carefully chosen. To do this, a full factorial experiment was designed with 24 treatments. The levels for C_w were set at 0.97 and 0.99, for C_g at 0.85, 0.9, and 0.95, and for $nABS$ at 0, 0.2, 0.4, and 0.6, based on our preliminary testing. The algorithm's stopping condition was a time limit of 60 seconds, with a population size of 100. Each treatment was tested in 10 replicates, resulting in a total of 240 trials. These trials were randomly conducted on the second problem within each of the six datasets. The average *IGD* was measured, and a significance level (α) of 0.05 was used for the analysis. The true PF for each problem was estimated by selecting the nondominated solutions from all 240 trials for that problem. To ensure comparability, the *IGD* values obtained from each trial were normalized to a range of [0,1] using min-max normalization. The results of the ANOVA are presented in the following Table 6.

In our ANOVA, distinct patterns emerged in the behavior of the *IGD*

performance metric across the different levels of the parameters C_w , C_g , and $nABS$. Significant differences were found among the different levels of each parameter— C_w , C_g , and $nABS$ —indicating that *IGD* is sensitive to variations in these parameters. This underscores the importance of carefully selecting their optimal values.

Fig. 3 presents the main effects plots for the *IGD* metric with respect to C_w , C_g , and $nABS$. Each plot illustrates how the average *IGD* changes as the levels of these parameters are varied. In the first plot, an increase in C_w from 0.97 to 0.99 leads to a slight reduction in the average *IGD*, suggesting a minor inverse relationship between C_w and *IGD*. The second plot shows that as C_g increases from 0.85 to 0.95, the average *IGD* decreases more noticeably, indicating a stronger inverse relationship between C_g and *IGD*. In the third plot, when $nABS$ increases from 0 to 0.2, the *IGD* drops sharply. However, as $nABS$ continues to increase from 0.2 to 0.6, the *IGD* slowly rises. This suggests that $nABS=0.2$ is the optimal value for minimizing *IGD*, and increasing $nABS$ beyond this point leads to a gradual deterioration in performance. Overall, the plots indicate that the optimal parameter combination for both *IGD* is $(C_w, C_g, nABS)=(0.99, 0.95, 0.2)$, as this set provides the best performance for *IGD*.

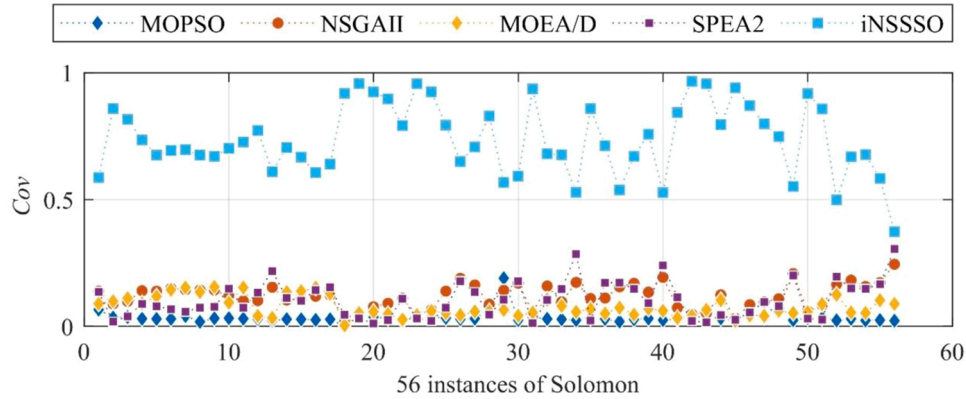
5.3. Comparison results

To conduct a comprehensive comparison and analysis of the performance of the proposed algorithm with the best setting determined in Section 5.3, our experimental comparison is chiefly divided into three segments: First, because only the transportation cost aligns with the objectives of past research, the iNSSSO is compared against advanced algorithms from the literature that solve the VRPTW based only on transportation cost as the objective value of the MO-VRPTW. Second, to justify the efficiency and effectiveness of iNSSSO, we use a suite of multi-objective performance metrics and compare them with other MOEAs.

Table 16

Performance results of the Solomon's RC2 instances.

Data	Criteria	MOPSO	NSGAII	MOEA/D	SPEA2	NSSSO	iNSSSO
RC201	Cov	0.02(0)	0.21(0.09)	0.05(0.02)	0.2(0.08)	0.04(0.04)	0.55(0.06)
	IGD	380.96(48.73)	90.78(34.08)	100.86(74.77)	99.34(64.6)	930.66(134.57)	36.17(11.95)
	HV	0.11(0.07)	0.39(0.04)	0.25(0.06)	0.39(0.04)	0.21(0.05)	0.44(0.04)
	Nnds	53.8(15.47)	99.53(0.99)	51.13(11.23)	99.93(0.26)	95.73(5.23)	99.73(1.03)
RC202	Cov	0.03(0.01)	0.06(0.02)	0.06(0.02)	0.03(0.02)	0(0.01)	0.92(0.04)
	IGD	317.32(80.65)	63.75(11.75)	109.01(39.79)	101.98(18.25)	1230.21(94.39)	7.7(5.2)
	HV	0.07(0.02)	0.27(0.03)	0.15(0.02)	0.27(0.03)	0.06(0.02)	0.42(0.02)
	Nnds	49.4(10.16)	100(0)	49.07(8.68)	99.73(1.03)	99.87(0.52)	100(0)
RC203	Cov	0.03(0)	0.08(0.04)	0.09(0.04)	0.03(0.03)	0.01(0.01)	0.86(0.06)
	IGD	462.45(81.85)	69.6(9.97)	127.18(94.72)	112.47(25.57)	1354.57(145.8)	14.13(15.24)
	HV	0.12(0.03)	0.32(0.05)	0.21(0.03)	0.31(0.06)	0.07(0.05)	0.47(0.04)
	Nnds	18(5.46)	99.93(0.26)	64(13.33)	99.93(0.26)	99.93(0.26)	99.93(0.26)
RC204	Cov	0.02(0)	0.17(0.06)	0.12(0.04)	0.2(0.08)	0.06(0.04)	0.5(0.07)
	IGD	334.81(43.42)	82.45(48.17)	104.9(22.91)	219.94(111.43)	1102.94(153.17)	59.38(50.45)
	HV	0.18(0.03)	0.4(0.03)	0.31(0.03)	0.39(0.03)	0.21(0.03)	0.48(0.03)
	Nnds	82.07(16.99)	100(0)	99.73(0.59)	100(0)	100(0)	100(0)
RC205	Cov	0.03(0.01)	0.18(0.08)	0.05(0.02)	0.15(0.07)	0(0.01)	0.67(0.08)
	IGD	488.29(91.22)	45.22(9.44)	83.32(38.86)	60.82(22.59)	1115.26(100.15)	22.67(7.21)
	HV	0.11(0.03)	0.35(0.03)	0.2(0.03)	0.35(0.02)	0.09(0.04)	0.4(0.02)
	Nnds	7.67(2.26)	99.87(0.52)	58(10.34)	99.73(1.03)	99.93(0.26)	100(0)
RC206	Cov	0.02(0.01)	0.16(0.06)	0.05(0.03)	0.15(0.09)	0.01(0.02)	0.68(0.08)
	IGD	529.66(86.57)	51.6(13.46)	68.16(12.74)	97.52(38.58)	1041.94(105.92)	26.48(14.1)
	HV	0.1(0.02)	0.39(0.03)	0.25(0.02)	0.39(0.03)	0.17(0.06)	0.47(0.03)
	Nnds	7.73(3.28)	100(0)	76.4(11.64)	100(0)	100(0)	100(0)
RC207	Cov	0.02(0)	0.17(0.08)	0.1(0.05)	0.17(0.07)	0.02(0.03)	0.58(0.08)
	IGD	567.46(105.33)	48.79(16.85)	63.33(6.59)	116.82(34.11)	1080.49(145.06)	23.33(7.23)
	HV	0.11(0.03)	0.4(0.02)	0.28(0.03)	0.39(0.02)	0.2(0.02)	0.47(0.02)
	Nnds	27.27(8.13)	100(0)	96.93(4.61)	100(0)	99.73(1.03)	100(0)
RC208	Cov	0.02(0)	0.25(0.1)	0.09(0.04)	0.31(0.12)	0.03(0.02)	0.37(0.08)
	IGD	542.87(42.58)	54.38(23.68)	123.89(132.88)	113.78(45.86)	1220.62(122.44)	40.76(18.09)
	HV	0.13(0.03)	0.47(0.03)	0.31(0.04)	0.47(0.03)	0.22(0.05)	0.5(0.03)
	Nnds	43.93(8.71)	100(0)	90.87(19.09)	100(0)	100(0)	99.73(1.03)
Overall	Cov	0.04(0.05)	0.16(0.05)	0.07(0.02)	0.15(0.08)	0.02(0.02)	0.64(0.15)
	IGD	436.31(111.66)	60.77(14.88)	92.28(21.69)	109.33(39.09)	1027.17(192.57)	33.16(29.39)
	HV	0.13(0.06)	0.38(0.06)	0.26(0.06)	0.38(0.06)	0.16(0.06)	0.46(0.03)
	Nnds	38.21(25.26)	99.92(0.14)	76.06(19.24)	99.94(0.1)	98.06(5.24)	99.91(0.1)

**Fig. 5.** Cov comparison of algorithms on the Solomon benchmark.

Finally, to further investigate the versatility and applicability of ABS, the local search mechanism integrated into iNSSSO, we analyze its impact by incorporating it into other well-established algorithms.

5.3.1. Comparison with previous results

In this comparison, we evaluate the effectiveness of the iNSSSO algorithm in minimizing transportation costs for the VRPTW. The performance of iNSSSO is compared with other leading solutions, including the best-known results obtained from the Solomon benchmark dataset (<http://web.cba.neu.edu/~msolomon/heuristi.htm>), as well as solutions from MODLEM[16], HSPEA2[43], and N-CLPSO[44]. The result as detailed in Tables 7, 8 and 9, highlighting instances where iNSSSO matches or surpasses the advanced algorithms, particularly in instances

R101, R201 –203, R209 –,210, RC201 –203, and RC205 –RC206. Bold figures indicate where iNSSSO excels, emphasizing its cost-reduction capabilities. The algorithm's effectiveness is further supported by its superior average rankings—2.76 in Tables 7, 3.04 in Table 8, and 2.75 in Table 9—underscoring its proficiency in addressing complex routing problems. These findings demonstrate iNSSSO's outstanding performance in optimizing transportation costs.

5.3.2. Comparison with other MOEAs

The efficacy of the proposed iNSSSO algorithm is further evaluated in comparison with established MOEAs. Given that the MO-VRPTW represents a novel problem formulation, with no MOEAs specifically tailored for it, selecting appropriate benchmark algorithms posed a

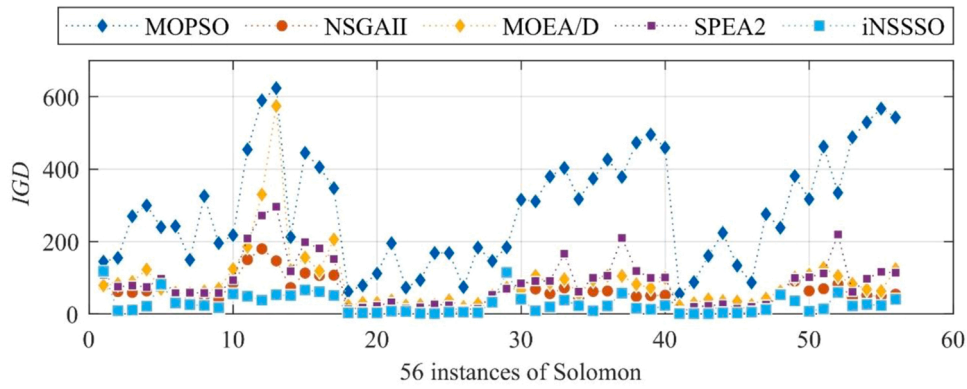


Fig. 6. IGD comparison of algorithms on the Solomon benchmark.

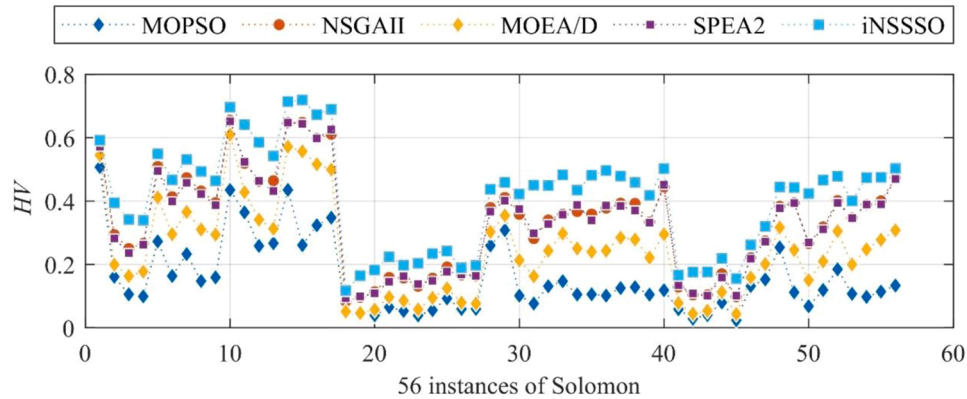


Fig. 7. HV comparison of algorithms on the Solomon benchmark.

unique challenge. Therefore, this study compares the performance of iNSSSO against widely recognized algorithms, including MOPSO, NSGA-II, MOEA/D, SPEA2, and the original NSSSO. These algorithms were chosen due to their proven effectiveness in addressing a wide range of multi-objective problems (MOPs) [46–48], with all except NSSSO sourced from <https://yarpiz.com/>.

The algorithms NSGA-II, MOEA/D, and SPEA2 utilized a segment-based crossover strategy, complemented by a uniform mutation. In addition to the velocity and position update mechanism, MOPSO also employed uniform mutation as an auxiliary operation [49]. The parameter settings for each algorithm were optimized in the same manner as iNSSSO in Section 5.2, using a full-factorial experimental design. For each algorithm, three levels were defined for the key parameters, including crossover probability (P_c), mutation probability (P_m), and mutation rate (R_m). For NSGA-II, MOEA/D, and SPEA2, the mutation probability was set as $1-P_c$. For MOEA/D, the number of weight divisions (T) was also considered a key parameter. The archive size for both MOEA/D and SPEA2 was uniformly set at 100. The range of parameter levels, as outlined in Table 10, was carefully selected based on literature recommendations and preliminary experimental results. A full-factorial experiment was subsequently conducted, testing all possible parameter combinations. Main effects analysis was used to assess the influence of each parameter on algorithm performance, with the results displayed in Fig. 4

Fig. 4. Main effects plot for the selected MOEAs. Finally, the optimal parameter combinations, which most effectively enhanced algorithm performance, are highlighted in bold in Table 10 for use in subsequent comparisons.

In the conducted comparative analysis, each MOEA, including MOPSO, NSGA-II, MOEA/D, SPEA2, NSSSO, and iNSSSO, was executed 15 independent times for each problem set, with a population size fixed

at 100. All algorithms were run until a predetermined termination criterion of 60 seconds was reached. NSSSO did not use the LKH initialization mechanism or ABS, while MOPSO, NSGA-II, MOEA/D, and SPEA2 employed only the LKH initialization mechanism. These methods were compared against the full approach proposed in this study, iNSSSO, which combines NSSSO with the LKH initialization mechanism and ABS. The true Pareto solution set (P) for each problem was accurately estimated by aggregating the final nondominated sets from all algorithms across all runs and then selecting the nondominated solutions from this cumulative set. These final nondominated solutions represent the comprehensive set of optimal solutions identified by the MOEAs over the course of the 15 runs.

The computational findings from the Solomon's benchmark tests, including the C1, C2, R1, R2, RC1, and RC2 series, are compiled in Table 11–16 and illustrated in Fig. 5–8. These tables and visuals compare the performance of the algorithms based on criteria such as Cov , IGD , HV , and the number of nondominated solutions ($Nnds$). Specifically, Fig. 5 illustrates the Cov comparison, Fig. 6 depicts the IGD values, and Fig. 7 outlines the HV scores across all algorithms on the Solomon benchmark. The results in Table 11–16 enumerate the mean and the standard deviation (presented in parentheses) of these performance metrics, aggregated from the 15 runs. The most advantageous mean values for each criterion across each problem are highlighted in bold. The ensuing synthesis of these results yields several pertinent insights:

1. In most cases, iNSSSO consistently shows the best performance across different metrics, particularly in terms of IGD and $Nnds$, which indicate better proximity to the true Pareto front and the diversity of nondominated solutions. This suggests that the integration of both

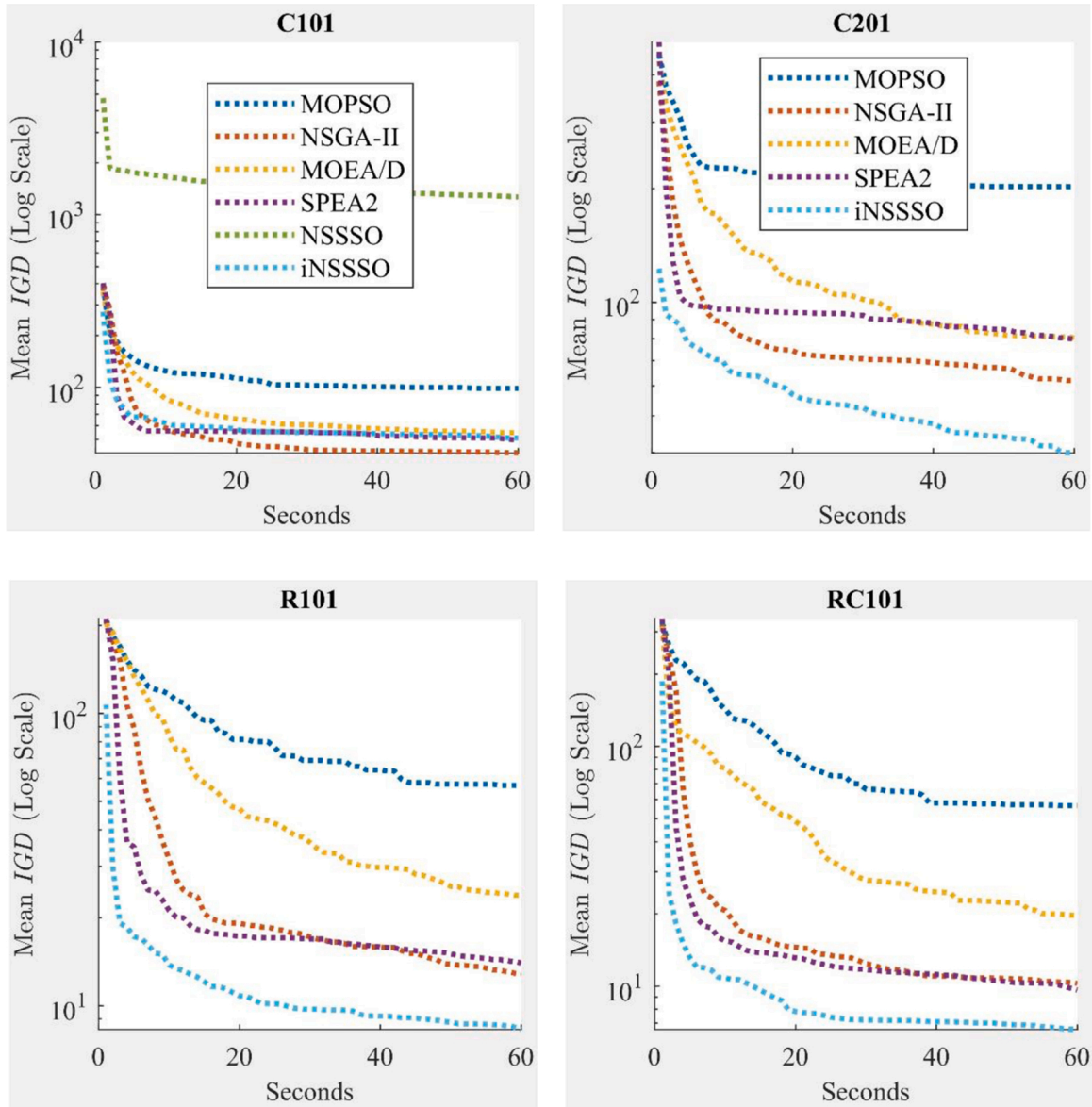


Fig. 8. Convergence performance of algorithms on four benchmark problems.

- LKH initialization and the ABS mechanism significantly enhances iNSSSO's performance over NSSSO and other methods.
2. NSSSO generally performs worse compared to iNSSSO, especially in cases where the problem complexity increases (e.g., C2 and R2 series). Without LKH initialization and ABS, its performance is outmatched in terms of *IGD* and *HV*, highlighting the importance of these mechanisms.
 3. Among the standard MOEAs (MOPSO, NSGA-II, MOEA/D, SPEA2), none consistently outperform iNSSSO in all the tested instances. For example, in C1 and R1 series, SPEA2 and MOEA/D show comparable performance in terms of *HV*, but iNSSSO remains superior in *IGD* and *Nnds*, indicating its balanced performance in both convergence and solution diversity. This superior performance of iNSSSO is largely attributed to the inclusion of the ABS mechanism, which enhances local search capabilities.
 4. In some problem instances, such as C205 or R109, the *IGD* and *HV* metrics fluctuate across algorithms, but iNSSSO remains within the top-performing methods. This suggests that while other algorithms may occasionally converge to the Pareto front, iNSSSO maintains a more consistent performance across various problem instances.

5. Based on Figs. 5–8, it is clear that iNSSSO outperforms other mainstream MOEA algorithms in both solution quality and diversity. This demonstrates that the combination of the ABS mechanism and LKH initialization enhances iNSSSO's performance in different ways. The LKH initialization improves the quality of the initial solutions, accelerating the early convergence process, while the ABS mechanism strengthens local search capabilities, allowing iNSSSO to escape local optima and further refine solutions.

In Fig. 8, there are four *IGD* convergence plots. In the top-left plot for the C101 problem, the NSSSO algorithm is included to demonstrate the effect of not using the LKH initialization mechanism. The plot shows that NSSSO's *IGD* convergence is noticeably slower than that of the algorithms utilizing LKH initialization, highlighting the importance of the LKH mechanism in accelerating the convergence process. The other three plots exclude NSSSO to further compare the performance with and without the ABS mechanism. For the C201 problem, iNSSSO exhibits the fastest convergence among all algorithms, and its final *IGD* value is the lowest, demonstrating its stable optimization capability. In the R101 and RC101 problems, the results similarly show that iNSSSO's convergence curve is steeper than those of the other algorithms, especially

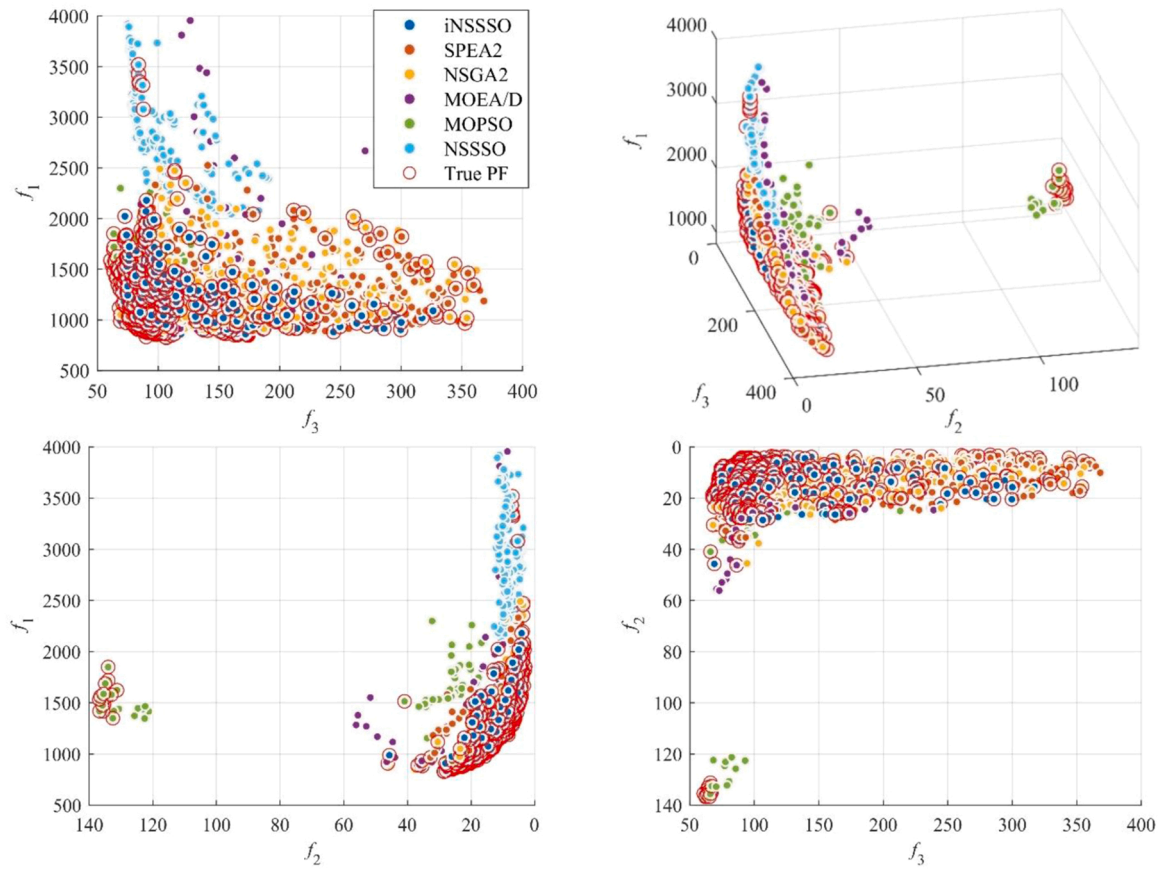


Fig. 9. Aggregate final nondominated solution set from all considered algorithms for C101.

Table 17
Improvement rates of performance metrics after integrating ABS (%).

Data	Criteria	iMOPSO	iNSGAI	iMOEAD	iSPEA2	iNSSSO
C1	Cov	20.64	269	3.94	1583.2	305.42
	IGD	102.23	27.04	7.29	40.27	74.93
	HV	77.5	16.04	16.4	20.95	20.5
	Nnds	101.14	−0.27	19.04	−0.48	0.11
C2	Cov	18.86	259.05	−4.56	1673.8	323.74
	IGD	19.44	45.88	22.64	52.16	77.79
	HV	39.74	14.79	8.93	13.54	18.64
	Nnds	44.44	−1.89	−4.6	−0.22	−0.16
C3	Cov	38.13	300.31	183.91	731.53	442.62
	IGD	166.58	24.18	9.54	35.78	29.74
	HV	86.19	22.16	47.21	26.11	29.15
	Nnds	97.71	3.61	77.65	0.25	5.65
C4	Cov	30.51	236.58	102.6	886.36	346.41
	IGD	116.13	24.32	−6.02	41.15	80.42
	HV	143.19	21.88	26.19	23.75	32.07
	Nnds	92.39	−0.05	−3.7	0.04	−0.02
C5	Cov	38.88	358.65	250.48	876.82	447.16
	IGD	113.79	12.21	−9	22.92	37.9
	HV	80.17	18.79	28.45	20.04	26.98
	Nnds	101.58	0.76	51.51	0.03	4.11
C6	Cov	23.47	241.88	117.76	620.68	335.38
	IGD	124.44	8.88	−15.81	30.85	62.52
	HV	130.88	18.05	21.41	19.39	27.74
	Nnds	138.38	−0.04	9.09	0.06	−0.03
Overall	Cov	28.42	277.58	109.02	1062.07	366.79
	IGD	107.10	23.75	1.44	37.19	60.55
	HV	92.95	18.62	24.77	20.63	25.85
	Nnds	95.94	0.35	24.83	−0.05	1.61

within the first 30 seconds, where the solution quality improves significantly faster than the algorithms without the ABS mechanism. This indicates that the ABS mechanism plays a critical role in accelerating

local search and improving early convergence performance.

In Fig. 9, the aggregate final nondominated solution set for the C101 problem is presented, including the solutions found by all the considered algorithms. The red circles represent the true PF. The scatter plots in different dimensions show how each algorithm's solutions are distributed across the objective space. From the plots, it is evident that iNSSSO (indicated in blue) produces solutions that are closer to the true Pareto front across all objective dimensions, demonstrating its superior ability to find high-quality nondominated solutions. In contrast, NSSSO (indicated in cyan), which does not utilize the LKH initialization or ABS mechanism, generates solutions that are more scattered and farther from the true Pareto front. The other algorithms, such as SPEA2, MOEA/D, and NSGA-II, show some proximity to the Pareto front but are generally less concentrated than iNSSSO.

5.3.3. Performance analysis of ABS integration into compared algorithms

To further evaluate the effectiveness of the proposed ABS, this section explores its performance when integrated into other algorithms. Embedding ABS into the compared algorithms is intended to determine whether its performance can be enhanced across benchmark instances. Experimental results are presented to highlight the comparative advantages of the algorithms before and after modification, thereby validating the applicability of the ABS mechanism in different algorithmic frameworks.

All compared algorithms were combined with ABS and re-solved using Solomon's benchmark datasets. Table 18 presents the overall performance results of compared algorithms on Solomon's dataset, both with (indicated by the prefix "i") and without ABS. To more clearly illustrate the improvement of each algorithm after incorporating ABS, the improvement rates for Cov, Hv, and Nnds are calculated as and for IGD is calculated as −.

Table 18
Comparative Performance of Original and ABS-Enhanced Algorithms on Solomon's Dataset.

Data	Criteria	MOPSO	iMOPSO	NSGAI	iNSGAI	MOEA/D	iMOEA/D	SPEA2	iSPEA2	NSSSO	iNSSSO
C1	Cov	0.03(0.01)	0.04(0.01)	0.06(0.01)	0.23(0.02)	0.09(0.02)	0.09(0.03)	0.02(0.01)	0.3(0.05)	0.08(0.03)	0.32(0.06)
	IGD	216.53 (64.11)	107.07 (36.51)	47.13 (11.97)	37.09 (7.36)	68.22(18.74)	63.59(12.1)	63.77(17.3)	45.46(8.05)	45.98 (17.21)	26.28 (3.26)
	HV	0.14(0.07)	0.25(0.06)	0.36(0.09)	0.41(0.07)	0.26(0.08)	0.3(0.05)	0.35(0.08)	0.42(0.07)	0.35(0.09)	0.42(0.07)
	Nnds	28.49(10.91)	57.3(5.73)	99.92(0.15)	99.64 (0.28)	71.7(12.67)	85.36(9.86)	99.93 (0.12)	99.44(0.68)	99.82(0.24)	99.93 (0.12)
C2	Cov	0.04(0.01)	0.05(0.01)	0.07(0.01)	0.24(0.03)	0.09(0.04)	0.09(0.03)	0.02(0.01)	0.32(0.05)	0.07(0.02)	0.3(0.05)
	IGD	400.23 (171.74)	335.09 (120.09)	120.85 (52.69)	82.84 (23.65)	179.51 (117.32)	146.38 (46.23)	171.92 (66.35)	112.99 (49.71)	127.22 (59.42)	71.56 (22.25)
	HV	0.26(0.09)	0.36(0.09)	0.54(0.1)	0.61(0.09)	0.45(0.12)	0.49(0.1)	0.53(0.1)	0.61(0.08)	0.52(0.1)	0.62(0.09)
	Nnds	40.8(18.5)	58.93(16.05)	99.81(0.43)	97.93 (4.83)	94.37(4.02)	90.03 (11.72)	99.86 (0.26)	99.64(0.4)	99.73(0.73)	99.57 (0.69)
R1	Cov	0.04(0.01)	0.05(0.03)	0.06(0.02)	0.23(0.03)	0.05(0.02)	0.13(0.06)	0.03(0.02)	0.25(0.04)	0.07(0.02)	0.37(0.05)
	IGD	115.8(39.32)	43.44(17.98)	16.49(9.92)	13.28 (12.08)	28.47(8.73)	25.99 (14.17)	22.34(9.99)	16.45 (16.05)	14.07(5.3)	10.84 (15.06)
	HV	0.08(0.08)	0.16(0.09)	0.18(0.1)	0.22(0.09)	0.12(0.1)	0.17(0.09)	0.18(0.1)	0.22(0.09)	0.18(0.1)	0.23(0.1)
	Nnds	24.23(11.52)	47.9(11.17)	95.52 (12.91)	98.96 (1.66)	49.4(18.47)	87.76(13.7)	99.47(0.93)	99.72(0.41)	94.49 (15.48)	99.83 (0.39)
R2	Cov	0.04(0.01)	0.05(0.02)	0.07(0.02)	0.25(0.04)	0.06(0.01)	0.12(0.04)	0.03(0.03)	0.29(0.04)	0.07(0.01)	0.3(0.07)
	IGD	406.37 (90.47)	188.02 (75.33)	55.56 (13.11)	44.69 (12.63)	82.28(27.71)	87.56 (27.94)	92.41 (29.64)	65.47 (24.73)	56.08 (19.68)	31.08 (11.7)
	HV	0.11(0.06)	0.27(0.04)	0.35(0.05)	0.43(0.04)	0.24(0.05)	0.31(0.05)	0.35(0.05)	0.43(0.03)	0.34(0.05)	0.44(0.03)
	Nnds	25.19(19.2)	48.46(8.99)	99.94(0.1)	99.89 (0.29)	81.44(15.69)	78.43 (18.02)	99.87(0.29)	99.91(0.18)	99.91(0.27)	99.89 (0.12)
RC1	Cov	0.03(0.01)	0.05(0.02)	0.06(0.02)	0.25(0.04)	0.05(0.01)	0.16(0.06)	0.02(0.02)	0.24(0.03)	0.06(0.01)	0.32(0.05)
	IGD	224.4 (168.58)	104.96 (89.66)	25.81 (16.24)	23(17.35)	41.77(21.04)	45.9(20.95)	43.39(32.2)	35.3(28.26)	24.13 (15.92)	17.5 (16.39)
	HV	0.11(0.09)	0.19(0.1)	0.25(0.13)	0.29(0.13)	0.17(0.11)	0.22(0.11)	0.24(0.13)	0.29(0.13)	0.24(0.13)	0.3(0.13)
	Nnds	26.81(14.56)	54.03(11.62)	99.02(2.33)	99.78 (0.35)	58.88(22.77)	89.21 (10.05)	99.83(0.2)	99.86(0.24)	95.95 (10.57)	99.89 (0.14)
RC2	Cov	0.04(0)	0.05(0.02)	0.07(0.02)	0.24(0.03)	0.06(0.02)	0.14(0.04)	0.04(0.03)	0.27(0.02)	0.07(0.01)	0.29(0.06)
	IGD	441.46 (115.82)	196.69 (76.25)	51.42(7.45)	47.23 (8.69)	71.22(16.81)	84.6(24.9)	93.8(28.1)	71.68 (18.15)	48.79 (13.65)	30.02 (11.08)
	HV	0.11(0.06)	0.26(0.05)	0.36(0.06)	0.43(0.04)	0.25(0.06)	0.3(0.05)	0.36(0.06)	0.43(0.04)	0.35(0.05)	0.44(0.03)
	Nnds	21.78(16.9)	51.93(9.73)	99.91(0.13)	99.87 (0.17)	79.07(18.64)	86.25 (11.21)	99.86(0.25)	99.92(0.12)	99.96 (0.06)	99.93 (0.12)

Table 19
Results of the statistical test on Cov.

Method	Friedman's test			Holm's test		
	Rank	Statistic	p-value	Statistic	p-value	APV
MOPSO	4.86	224.71	< 0.001	10.91	< 0.001	< 0.001
NSGAI	2.52			4.29	< 0.001	< 0.001
MOEA/D	3.45			6.92	< 0.001	< 0.001
SPEA2	3.45			6.92	< 0.001	< 0.001
NSSSO	5.73			13.38	< 0.001	< 0.001
iNSSSO	1.00			0.00	< 0.001	

Table 20
Results of the statistical test on IGD.

Method	Friedman's test			Holm's test		
	Rank	Statistic	p-value	Statistic	p-value	APV
MOPSO	5.00	260.99	< 0.001	10.91	< 0.001	< 0.001
NSGAI	2.02			2.47	< 0.001	< 0.001
MOEA/D	3.57			6.87	< 0.001	< 0.001
SPEA2	3.27			6.01	< 0.001	< 0.001
NSSSO	6.00			13.74	< 0.001	< 0.001
iNSSSO	1.14			0.00	< 0.001	

The improvement rate results in Table 17 show that the effectiveness of integrating ABS varies across different algorithms. In terms of overall performance, iSPEA2 achieved the highest improvement rate in the Cov metric, while iMOPSO demonstrated the best improvement rates in IGD, HV, and Nnds. Among all algorithms, the lowest improvement rates for Cov, IGD, and HV were 28.42 %, 1.44 %, and 18.62 %, respectively,

Table 21
Results of the statistical test on HV.

Method	Friedman's test			Holm's test		
	Rank	Statistic	p-value	Statistic	p-value	APV
MOPSO	5.30	265.30	< 0.001	12.17	< 0.001	< 0.001
NSGAI	2.32			3.74	< 0.001	< 0.001
MOEA/D	4.02			8.54	< 0.001	< 0.001
SPEA2	2.68			4.75	< 0.001	< 0.001
NSSSO	5.68			13.23	< 0.001	< 0.001
iNSSSO	1.00			0.00	< 0.001	

while the highest improvement rates were 1062.07 %, 107.10 %, and 92.95 %, respectively. In contrast, Nnds showed the smallest improvement among the four metrics, with iSPEA2 having an improvement rate of −0.05 %, and NSGAI and iNSSSO both achieving rates below 2 %. Based on the results in Table 18, it can be inferred that these algorithms already exhibit strong capabilities in searching for non-dominated solutions, limiting the additional benefits brought by ABS. Nevertheless, the overall results confirm that ABS provides stable and reliable performance improvements across different algorithms.

Table 18 summarizes the overall performance metrics of each algorithm across all problem instances. The results consistently show that iNSSSO outperforms all other algorithms across all metrics. These findings are consistent with the experimental results reported in Section 5.3.2, indicating that ABS is specifically designed to complement NSSSO's update mechanism. While ABS enhances the performance of other algorithms, its improvements are less significant compared to the advantages achieved by iNSSSO.

5.4. Computational Complexity analysis of iNSSSO

The primary computational costs in iNSSSO are composed of three key components: non-dominated sorting, crowding distance calculation, and solution updating. The non-dominated sorting uses the divide and conquer algorithm, with a time complexity of $O(N \log^{M-1} N)$ in the best case and $O(N^2 \log^{M-2} N)$ in the worst case [36], where M represents the number of objectives and N denotes the population size. Crowding distance calculation is employed to ensure the diversity of solutions, and its time complexity is $O(MN \log N)$, due to the need to sort solutions based on objective values. The solution update process, applied across the entire population, incurs a time complexity of $O(N)$. Therefore, the overall time complexity of iNSSSO can be expressed as $O(N^2 \log^{M-2} N)$ in the worst case, dominated by the nondominated sorting phase, particularly for large population size and higher dimensional objective spaces.

5.5. Nonparametric statistical test

In this section, a nonparametric statistical test was applied to the results discussed in Section 5.3.2 to critically evaluate the performance differences among the considered MOEAs [50]. The statistical findings related to *Cov*, *IGD*, and *HV* are detailed in Tables 19, 20, and 21, respectively. Using Friedman's test to examine the comparative outcomes across algorithms, we observed p -values significantly lower than the alpha threshold of 0.05, leading to the rejection of the null hypothesis and affirming that distinct performance variations exist among the algorithms. To further substantiate these results, Holm's post hoc test was conducted, focusing on pairwise comparisons. The analysis presents the adjusted p -value (APV), a more rigorous measure than the standard p -value, underscoring the performance disparity between iNSSSO and its competitors. This detailed evaluation demonstrates that iNSSSO significantly outperforms in metrics such as *Cov*, *IGD*, and *HV*, as detailed in the referenced tables.

These statistical examinations corroborate the superior performance of iNSSSO, as initially reported in Section 5.3.2. The compelling evidence from the nonparametric tests firmly establishes iNSSSO as a leading algorithm in *Cov*, *IGD*, and *HV* measurements, surpassing benchmarks set by established algorithms like MOPSO, NSGA-II, MOEA/D, and SPEA2. Therefore, the experimental outcomes from Section 5.3.2 are convincingly validated by this statistical analysis, reinforcing iNSSSO's effectiveness and potential in addressing MO-VRPTW problems.

6. Conclusion and future work

This research advances the MO-VRPTW resolution by harmonizing three pivotal objectives: reducing travel distance, bolstering customer contentment, and distributing driver workload evenly to serve the multifaceted interests of the supply chain's principal entities. The proposed iNSSSO algorithm is central to this enhancement, integrating LKH for initial high-quality solutions and the ABS for refined local searches, fostering both swift convergence and expansive solution diversity.

iNSSSO's success emanates from its deft initiation and progressive refinement process. The LKH propels the algorithm toward promising solution territories, while ABS, informed by A* algorithm principles, sharpens the exploitation in the search space. This dual strategy not only facilitates accelerated convergence to quality solutions but also ensures a judicious equilibrium between exhaustive exploration and focused exploitation—enabling the discovery of varied solutions and the deepened pursuit of viable regions. Empirical assessments affirm iNSSSO's superior efficacy over algorithms like NSSSO, MOPSO NSGA-II, MOEA/D, and SPEA2, as statistically validated by Friedman's and Holm's tests.

However, the proposed iNSSSO algorithm is specifically designed

and fine-tuned for the VRPTW problem. The initialization mechanism and ABS-based local search are tailored for this context, which may limit its applicability to other types of multi-objective optimization problems. Therefore, applying the iNSSSO framework to real-world VRPTW scenarios presents an important direction for future work, where further investigation may refine both the initialization and local search strategies to better address the complexities of practical applications. Moreover, in this study, using transportation cost as one of the objectives may not always be practical, as increasing costs to balance workload does not always align with business logic. A potential alternative research direction could involve incorporating makespan as an objective or introducing a spread-time constraint (e.g., requiring vehicles to return to the depot within a specified time). These adjustments may more accurately reflect real-world operational priorities and improve the algorithm's applicability in practical scenarios.

CRedit authorship contribution statement

Chiu Chun-Chih: Writing – review & editing, Validation, Methodology. **Chen Tzu-Li:** Writing – review & editing, Validation. **Lai Chyh-Ming:** Writing – original draft, Validation, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research received funding from the National Science and Technology Council of the Republic of China under contract number NSTC 112-2221-E-606-013.

Data availability

The authors do not have permission to share data.

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